SET 1

1. What do you call the simultaneous two-way mode of communication?
   A. full duplex C. half duplex
   B. simplex D. full-full duplex
2. Which of the following is represented by CMOS?
   A. Complementary Metal Oxide Semiconductor Logic
   B. Complimentary Mutual Oxide Semiconductor
   C. Common Oriented Business Logic
   D. Complementary Metal Oxide Semiconductor
3. Which of the following represented by COBOL?
   A. Common Business Language
   B. Common Business Oriented Language
   C. Common Oriented Business Language
   D. Commun Business Orientation Language
4. Cellular phone operates on which frequency?
   A. VHF C. EHF
   B. UHF D. SHF
5. Channel 2 to 13 operates on which frequency?
   A. VHF C. EHF
   B. UHF D. SHF
6. When a fluid flow through a pipe of cross sectional are A with a velocity v, the discharge is
   A. Av^2 C. A^2V
   B. Av D. A^2/2v
7. Processing system is a processor limiter when sorting of small table in memory would speed up computation?
   A. Adding Cache memory
   B. Adding more Main memory
   C. Adding Virtual memory
   D. Adding Peripheral memory
8. What allows choosing command from men?
   A. files
   B. DIR
   C. storage
   D. Menu Drive Interface
9. Which of the following is a member of the Halogen family?
   A. sodium C. phospliorgas
   B. fluorine D. hydrogen
10. In fluid flow, if the fluid travels parallel to the adjacent layers and the paths of Individual particles do not cross, the flow is
    A. laminar C. dynamic
    B. turbulent D. uniform
11. The state of thermodynamic system is always defined by its
    A. process
    B. availability
    C. properties
    D. temperature and pressure
12. Compared to gases and liquids, solid are less
    A. color C. weight
    B. reactive D. heat
13. Enthalpy of an ideal gas is a function of
    A. internal energy C. temperature
    B. entropy D. pressure
14. Entropy is the measure of
    A. The change in enthalpy of a system
    B. The internal energy of a gas
    C. The heat capacity of a substance
    D. randomness or disorder
15. When the output of the AND gate is inverted which gate is formed?
    A. OR C. NAND
    B. NOR D. DNA
16. Nibble is composed of how many bits?
    A. 4 C. 6
    B. 8 D. 7
17. What is the actual shape of the methane CH, molecule?
    A. Tetrahedral C. square planar
    B. linear D. pyramid
18. Reactions generally proceed faster at higher temperature because of which of the following
    A. the activations energy is less
    B. the molecule are less energetic
    C. the bonds are easier to break
    D. the molecules collides more frequently
19. Which of the following does not threatens the operation of the computer?
    A. Trojan horse C. worm
    B. firewall D. hacking
20. One Newton Is equivalent to
    A. 10^5 dyne C. 10^7 dyne
    B. 10^8 dyne D. 10^6 dyne
21. The radical ion (NO2) is called
    A. nitrite C. nitride
    B. nitrate D. binitrite
22. Which of the following metals has the highest specific heat at 100°C?
    A. aluminum C. copper
    B. bismuth D. iron
23. In a spreadsheet (EXCEL), the number in cell A2 is set to 2. Then cell B2 Is set to A2*4A2, where $ indicates the absolute cell address. This formula is copied into cells C2and D2. What Is the value shown in cell D2?
    A. 8 C. 4
    B. 2 D. 16
24. The circuit conductors between the final overcurrent device protecting the circuit and the outlet(s).
    A. Feeder C. Branch Circuit
    B. Service lateral D. Service Conductor
25. What is the old Electrical Engineering law?
    A. RA 814 C. RA 9136
    B. RA 184 D. RA 7920
26. What is another term for acid-test ratio?
    A. Current ratio
    B. Quick ratio
    C. Profit margin ratio
    D. Price-earnings ratio
27. A line to ground system, the white conductor that carries current in normal operation is?
    A. Neutral C. grounding conductor
    B. ground D. grounding electrode
28. A 4-wire system has a high-leg voltage that is indicated by a wire of color____
    A. Black C. brown
    B. orange D. red
29. What are considered as the "building blocks" for engineering materials?
    A. Atoms C. Matters
    B. Elements D. Compounds
30. Defects seen by optical or electron microscope
    A. Alteration C. atomization
    B. aberration D. deformation
31. Which is not used in programming applications?
    A. C C. Oracle
    B. Java D. Visual Basic
32. In the writing of PEC 2017, what is the version of National Electrical Code that was patterned to?
    A. 2008 C. 2001
    B. 2017 D. 2002
33. When a body is subjected to three mutually perpendicular stresses, of equal intensity, the ratio of direct stress to the corresponding volumetric strain is known as
    A. Young's modulus C. Bulk Modulus
    B. Modulus of Rigidity D. Poisson's ratio
34. What is the formula for a straight line depreciation rate?
    A. % net salvage value / estimated service life
    B. 100% net salvage value / estimated service life
    C. average net salvage value/ estimated service life
    D. (100%-% net salvage value) / estimated service life
35. Which of the following is represented by JPEG?
    A. Joint Photographic Experts Group
    B. Joint Photographic Experts Grade
    C. Joint Photography Experts Grade
    D. Joint Photography Experts Group
36. A circle in the flowchart symbols represent which one of the following?
    A. Connector C. Data
    B. Decision D. Terminator
37. A rectangular shape symbol in a flow chart represent,
    A. process C. merge
    B. decision D. preparation
38. FTTH mean,
    A. fiber to the home C. fiber to the hall
    B. fiber to the house D. filter to the home
39. Compare the metallic iron atom Fe and the ferric ions Fe+2, and Fe+3. Which has the smallest atomic radius at the same temperature?
    A. Fe+3 C. Fe
    B. Fe+2 D. they have same radius

40. What refers to the time with which a disc can find data being sought?
A. access time C. transfer time
B. data rate time D. enter time

41. What is the term for a high speed memory used to store a small amount of data temporarily so that the data can be retrieved quickly?
A. cache memory
B. video memory
C. volatile memory
D. scratchpad memory

42. The internal rate of return of a project that Involves an initial investment with subsequent positive cash flows is 18 percent. Four companies are considering the project. Given the following respective minimum attractive rated of return (MARR), which company will be most likely to accept the project?

Company	MARR
A	12 percent
B	14 percent
C	18 percent
D	19 percent

A. Company A C. Company C
B. Company B D. Company D

43. A diagram drawn to help visualize and simplify problem having diverse receipts and disbursements.
A Budget C. year-end-convention
B. investment D. Cash-flow

44. Which of the following relations defines enthalpy?
A. $H = u + \frac{p}{T}$ C. $h = u + \frac{p}{v}$
B. h=u+pv D. h=pV+T

45. What refers to chemically combined elements with definite proportions of the component elements?
A. Mixture C. Compound
B. Molecule D. Alloy

46. Which of the following cannot be a property of a gas?
A. Density C. Viscosity
B. Pressure D. Temperature

47. The relationship between the extension of a spring and the force required to produces the extension is
A. F=ma $C. F = \frac{mv2}{a}$
B. F= μ N D. F= kx

48. The place where buyers and sellers come together.
A. Market C. Recreation Center
B. Business D. Buy and sell section

49. The linear portion of the stress-strain diagram of steel is known as the
A. Modulus of elongation C. irreversible range
B. Plastic range D. elastic range

50. In any collision, which of the following is conserved?
A. Kinetic energy C. momentum
B. velocity D. potential energy

51. What is another term for modulus of elasticity?
A. Bulk modulus C. Constant of elongation
B. Shear strain D. Young's modulus

52. Which of the following is added to the drinking water distribution system for disinfection?
A Soda ash C. Lime
B. Chlorine D. Iodine`

53. An enclosed channel designed expressly for holding wires, cables, or busbars, with additional functions as permitted by the Philippine Electrical Code.
A. Duct C. Cable Tray
B. Raceway D. Tube

54. Sec. 23 of Art III of RA 7920. Issuance of Citification of Registration. The professional is required to renew license every years upon payment of registration fees.
A. 1 C. 3
B. 2 D. 4

55. The property that characterizes a material's ability to be drawn into a wire is its
A. Flexibility C. Malleability
B. Rigidity D. Ductility

56. Which gas law states that the total volume occupied by a mixture of gasses is equal to the sum of their pure component volume constant temperature and pressure?
A. Amagat C. Graham
B. Dalton D. Charles

57. Which of the following instruments for measuring fluid velocities does not need to be calibrated?
A. barometer C. Hot-Wire Anemometer
B. Pilot Tube D. Venturi Tube

58. Which of the following is not taken into account in making an economic decision between two alternative equipments?
A. Reliability C. Rate of return
B. Operating Cost D. Salvage value

59. Copper when exposed to ordinary atmosphere becomes oxidized turning it Into what color?
A. Brown C. Light orange
B. Light gray D. Black

60. The number of molecules in 22. 4 liters (under standard conditions) of a substance in Its gaseous state Is called,
A. Avogadro's number C. Petit's number
B. Dulong's number D. Graham's number

61. Aluminum hydroxide AI(OH) , Is called an amphoteric hydroxide because
A. it can react as either an acid or a base
B. it has limited solubility in water
C. it contains 3 hydroxyl groups
D. either one, two or all three hydroxyl group can ionize in water

62. The chemical formula of the salt formed from the neutralization of potassium hydroxide because
A. K2SO4 C. P2SO4
B. NaCI D. KSO4

63. Which of the following statements about the halogens is incorrect
A. they will react with water to form very strong base
B. they are good oxidizing agents
C. they all exhibit a valence a valence of -1
D. they form diatomic molecules

64. In a single phase, 3 wire electrical system, the middle conductor must be?
A. ungrounded C. hot
B. out of phase D. grounded

65. Which element is present in all organic compounds?
A. carbon C. oxygen
B. nitrogen D. phosphorus

66. The amount of lateral strain in a organic compounds?
A. the bulk modulus C. Poisson's ration
B. the yield stress D. Hooke's law

67. Which of the following cycle is used in vapour cycle power plant?
A. Brayton Cycle C. diesel cycle
B. Rankine Cycle D. Ericson cycle

68. vertically loaded beam, fixed at one end and simply supported at the other end is indetermine to what degree?
A. second C. third
B. first D. fourth

69. What mechanical property of material refers to the nominal stress at fracture in a certain test at constant load and constant temperature?
A. Creep strength
B. hardness
C. stress rupture strength
D. compressive yield strength

70. How many atoms in the unit cell of a body centered cubic (BCC) crystal structure?
A. 4 C. 2
B. 6 D. 8

71. In dwelling unit the lighting expressed as VA/sq m is at least,
A. 28 C. 16
B. 24 D. 8

72. Which conduit is not suitable for wet location?
A. RMC C. RNMC
B. IMC D. FMC (Flexible Metal Conduit)

73. What fraction of the full load current must be used es a factor to calculate the size of conductor for the capacitor?
A. 1/2 C. 1/4
B. 1/3 D. 1/5

74. It is a form of audio broadcasting on the internet.
A. webcasting C. newscasting
B. podcasting D. netcasting

75. Pycnometer is used to determined
A. water content and voids ratio
B. specific gravity and dry density
C. water content and specific gravity
D. no of these

76. The temperature at which vapor pressure of a liquids equal to the atmospheric pressure Is?
A. distillation C. boiling point
B. surface tension D. vaporization

77. Which of the following is used to find a particular webpage?
A. URL C. domain
B. IP address D. google

78. You are standing on a bathroom scale in a elevator. In which of this situations must the scale read the same as when the elevator is at rest?
A. Moving up at constant speed
B. Moving up at Increasing speed
C. Moving down with Increasing speed
D. In free fall (after the elevator cable has snapped)

79. What term is used to describe retrieval of data from a memory?
A. unloading C. reading
B. downloading D. accessing

80. The purpose of iron sulfide in steel alloying is to increase
A. atmospheric corrosion resistance
B. hardness and strength
C. corrosion resistance
D. brittleness

81. If disturbance is parallel to the direction of travel of a wave Is classified as
A. longitudinal C. transverse
B. electromagnetic D. torsional

82. What is the substance that cannot be decomposed into simpler substance during ordinary chemical reaction?
A. atom C. molecule
B. element D. compound

83. ___is a secondary storage medium, generally non removable disc made out of metal and covered with a magnetic recording surface. It holds data in the form of magnetized spots and are usually built in to the corapter system cabinet
A. hard disk C. compact disk
B. floppy disk D. CD ROM

84. Which of the following is the instrument used to measure to speed of a motor in pen?
A. tachometer C. speedometer
B. frequency meter D. pitot tube

85. Which of the following in the instrument used to measure the frequency?
A tachometer C. speedometer
B. frequency meter D. pitot tube

86. In PEC 1, rated voltage for computation of lighting receptacle and outlet is based on,
A. minimum C. effective
B. maximum D. average

87. What group of steels are high-carbon, high-chromium tools steels suitable for cold working applications?
A. Group A C. Group H
B. Group D D. Group M

88. In bathtub and shower areas no part of cord-connected fixtures, hanging fixtures, lightning track, pendants or celling-suspended (paddle) fans shall be located within a zone measured __ horizontally and __ vertically from the top of the bathtub rim or shower stall threshold.
A. 1000mm, 3000 mm C. 900mm, 2400mm
B. 600mm, 3000 mm D. 800 mm, 2500mm

89. A wire or Cable carrying electric current, supported by poles, tower of other structures, but not including vertical or lateral connecting wires.
A. Transmission Line C. Conductor
B. Line Conductor D. Lateral Conductor

90. Circuit breaker is a device designed to ____a circuit by nonautomatic means and to ___the circuit automatically on a predetermined overcurrent without damage to itself when property applied within the rating.
A. operate, disconnect C. close, open
B. disconnect, operate D. open, close

91. What refers to the value of an asset which a disinterested party, different from the buyer and seller, will determine in order to establish a price acceptable to both parties?
A. Book Value C. Fair value
B. Market Value D. Franchise value

92. The minimum diameter of a stainless-steel grounding electrode is
A. 10mm C. 14mm
B. 12mm D. 16mm

93. Two balls are thrown from the roof of a building with the same initial speed. One is thrown horizontally while the other is thrown at an angle of 20 degrees above the horizontal. Which hits the ground with the greatest speed? Ignore air resistance.
A. The one thrown horizontally
B. The one thrown at 20 degrees
C. They hit the ground with the same speed
D. The answer cannot be determined with the given Information

94. What is the principle works in the hydraulic brakes of an automobiles?
A. Bernoulli's law C. Archimede's principles
B. Pascal's law D. Posleullle's Law

95. What is the primary purpose of PEC?
A. to serve as guide for the untrained personnel
B. Is to conserve electrical energy
C. is the practical safeguarding of persons and property
D. to serve as manual guide for design

96. The process of outsmarting the search engine
A. Google fooling C. Search engine optimization
B. Rank Enhancement D. Search Engine Hacking

97. What percentage of the human body will float in fresh *water?* Average *density* of human body is 980 kg/ cu. m.
A. 98 C. 96
B. 2 D. 4

98. Which of the following CORRECTLY describes atoms located at gain boundaries?
A. They are located primarily in highly strained and distorted positions
B. All of these are correct
C. They have a highly free energy than atoms in the undisturbed part of the crystal lattice
D. They are subject to the same type of interatomic forces that are present in the interior atoms of the crystal.

99. For which of the following is the Program Evaluation and Review Technique (PERT) not used?
A. construction projects
B. preparation of bids and proposals
C. computer programing assignments
D. queueing problems

100. For a fluid, viscosity is defined as the constant of proportionality between shear stress and what other variable?
A. The spatial derivative of velocity
B. the time derivative of pressure
C. the time derivative of density
D. The spatial derivative of density

SET 2

1. GMC was created by the Energy Regulatory Commission to monitor the compliance of the uses of the Grid to the PGC. What does the GMC stands for?
   A. Grid Market Commission
   B. Generation Management Corporation
   C. Grid Management Committee
   D. Generator Management Committee

2. The customer is only charged or credited as the case may be, for the difference between the electricity supplied by the Distribution Utility and electricity it supplies to the Distribution Utility when the times It has excess VRE generation both of which are metered using either 2 unidirectional meters, one for Import and one for export, or a single bidirectional meter. What do you call this scheme?
   A. Market metering C. Import-export metering
   B. Net metering D. Dual metering

3. Which of the following statement is FALSE?
   A. kernel is the one of component of any os
   B. windows 98 is RTOS
   C. RTOS performance task in predictable amount of time
   D. Interrupts are use to develop RTOS

4. The government agency created under Presidential Decree 269, whose additional mandates includes preparing electric cooperative in operating and competing under a deregulated electricity market.
   A. National electrification Administration
   B. National electrification Commission
   C. National Grid Corporation
   D. National Power Corporation

5. What is PDC stands for In EPIRA?
   A. Philippine Distribution Commission
   B. Philippine Distribution Code
   C. Philippine Distribution Committee
   D. Philippine Distribution Coordination

6. Which of the following is/are the most important factor/s to be considered in the selection of fuel for power plants?
   I. Cost of fuel II. Calorific value of fuel
   A. I only C. II only
   B. Either I or II D. both I and II

7. RA 7832 concerns on_____
   A pilferage of electricity, theft of transmission materials and capping of system loss
   B. pilferage of electricity, theft of transmission materials and rationalizing of system loss
   C. pilferage of electricity energy
   D. pilferage of electricity and *electrical* materials

8. What governs the minimum safety requirement in the Philippines for distribution and grid?
   A. Philippine Electrical Safety Code
   B. Philippine Electrical code
   C. Philippine Grid Code
   D. Philippine Distribution Code

9. 34. 5 kV to 69 kV is classified as
   A. low voltage C. high voltage
   B. medium voltage D. extra high voltage

10. What kind of bond do you expect in bronze?
    A. metallic C. covalent
    B. Ionic D. Van Der Waal's

11. When peace and order problem exist, National Grid Corporation of the Philippines (NGCP) issues
    A. security red alert C. trouble alert
    B. imminent danger alert D. red alert

12. Which statement about area moments of inertia is false?
    A. The area moment of inertia arises whenever the magnitude of the surface force varies linearly with distance.
    B. The moment inertia of a large area is equal to the summation of the inertia of the smaller areas with in the large area.
    C. The areas closest to the ads of Interest are the most significant
    D. area moments of Inertia are always positive

13. Based on the Philippine Distribution Code, which of the following Interruption event is not excluded in the calculation of distribution utility reliability indices?
    A. outage that occur on 230 v lines
    B. outage that occur on 500 v lines
    C. outage that occur on 34. 5 kV lines
    D. outage that occur on 230 kV lines

14. Based on Philippine Distribution Code, retail competition and open access on distribution wire shall be implemented not later than three (3) years upon the effectively of the act, subjects to the following conditions, which Is not include?
    A. establishment of the wholesale electricity spot market
    B. Approval of unbundled transmission and distribution wheeling charges
    C. Initial Implementation of the cross subsidy removal scheme
    D. Non-privatization of the less 70 percent of the total capacity of generating assets of NPC Luzon and Visayas

15. The master control center for a cellular telephone system is the
    A. cell site
    B. mobile telephone switching office
    C. central office
    D. branch office

16. Which of the following principle will the operation of fiber-optic cable is based?
    A. dispersion C. reflection
    B. refraction D. absorption

17. Which of the following device is used in thermal power plants to reduce the level of pollution?
    A. pulverizer C. precipitator
    B. chimney D. Induced draft fan

18. When oxygen combines with an element to form a compound the resulting compound is called
    A. a salt C. oxidation
    B. an oxide D. an oxalate

19. The length of time usually assumed to be reversible?
    A. not in the choices C. receivable turnover
    B. payback period D. return on investment

20. Which of the following is usually assumed to be reversible?
    A. paddle wheel C. melting an ice cube
    B. compressing a gas D. a burst membrane

21. Spark plug makes use of which of the following materials for Insulation?
    A. porcelain C. ion
    B. proton D. anode

22. How would you plot multiple graphs in MATLAB?
    A. Using the hold function
    B. Using the mstem function
    C. Using the mplot function
    D. It cannot be done MATLAB

23. The most important element that controls the physical properties of steel is
    A. silicon C. manganese
    B. tungsten D. carbon

24. What is TCS in EPIRA stands for?
    A. Transition Supplier Contract
    B. Transition Supply Contract
    C. Transition Supplier Commission
    D. Transition Supply Commission

25. Which of the following is non-renewable energy?
    A. nuclear C. solar
    B. wind D. geothermal

26. Which of the following is not part of the requirements to take RME?
    A. taken 1-yr electrician course and 3-yrs apprenticeship
    B. taken 2-yr electrician course and 3-yrs apprenticeship
    C. taken 3-yr electrician course and 3-yrs apprenticeship
    D. graduate of BSEE

27. Two sky divers jump at the same time and want to play catch. Where should one aim if he wants the other to catch it
    A. below
    B. above because all falls faster
    C. above because ball falls slower
    D. straight because there's no air resistance

28. The purpose of boron in steel alloying is to Increase
    A. deoxidation C. hardness
    B. hardness and strength D. corrosion resistance

29. When engineers enters management, what is most likely problem he finds the difficult to acquire?
    A. Learning to trust others
    B. Learning how to work through others
    C. Learning how to take satisfaction in the work of others
    D. All of the above

30. Who is qualified to become a commissioner?
    A. Lawyer C. PEE
    B. PRBEE D. REE

31. It Is an Informal website publishing on the world wide web that consists of discrete often informal diary style text entries which are displayed in reverse chronological order so the most recent post appears first at the top of the web page.
A. blogging C. twitting
B. Instagram D. Facebook

32. What is the republic act number of the newly signed law of President Duterte called Murang Kuryente Act?
A. RA 11371 C. RA 10912
B. RA 9136 D. RA 7832

33. The radius of a bend is measured from the____degree point not Including the socket, through to the centerline of the conduit.
A. 180 C. 90
B. 270 D. 45

34. Equipment is delayed. What will you do?
A. Talk to team members to change activities
B. Report to manager
C. Replan
D. Let them notify the client about the delay

35. The Pentium is____bit microprocessor that was introduced by Intel on March 22, 1993 as the very first CPU In the Pentium brand.
A. 16 C. 64
B. 32 D. 86

36. Battery acid is the common name for
A. formic add C. nitric acid
B. hydrochloric acid D. sulfuric acid

37. ____Is the first computer designed and primarily used to calculate artillery firing tables for the United States Army Ballistic Research Laboratory
A. ENIAC C. APPLE
B. EDVAC D. Super Computer

38. Steels that are used for drills, cutting tools and lives are known as___carbon steel.
A. High C. Low
B. Medium D. Mild

39. How to print a handbook from Powerpoint Slides?
A. Open the Powerpoint presentation and dick on Print in the File Menu
B. Use the Export Feature in Powerpoint to create a PDF file and then print the pdf document
C. Copy and paste the slides into a word document and then print in word
D. Use the Save As feature in Powerpoint to create a word document and then print the word

40. The color of zinc Is:
A. greyish white C. greenish white
B. blush white D. yellowish white

41. What Is the ratio of the ultimate failure strain to yielding strain
A. Poisson's Ratio C. Ductility
B. Resilience D. Fatigue

42. Which of the following computer language is used for artificial Intelligence?
A. FORTAN C. C
B. PROLOG D. COBOL

43. The atmospheric pressure at sea level is
A. 103 Kn/m² C. 760 mm of mercury
B. 10. 3m of water D. All of the above

44. Property of a fluid by which its own molecules are attracted is called
A. Adhesion C. Viscosity
B. Cohesion D. Compressibility

45. In flow process, neglecting KE and PE changes, -VdP represents which Item
A. heat transfer C. dosed system work
B. shaft work D. flow energy

46. Which of the following Is spreadsheet software?
A. Microsoft Word C. Microsoft Excel
B. Notepad D. PowerPoint

47. Before starting my Installation work, alteration, repair or extension on any electrical system, what type of permit Is needed?
A. Building permit C. Electrical permit
B. Working permit D. Mayor's permit

48. A computer_____is self-replicating computer program
A. Viruses C. Spyware
B. Worms D. Malware

49. A glass of ice water is filed to the brim with water, the ice cubes stick up above the water surface. After the ice melts, which is true?
A. The water level is below the top of the glass.
B. The water level b at the top of the glass but no water has spilled.
C. Some water has spilled over the sides of the glass.
D. Impossible to say without knowing the initial densities of the water and the ice

50. For school buildings, the general lighting load is__VA per square meters.
A. 24 C. 16
D. 20 D. 8

51. Sec. 34 of Art. IV. Sundry Provisions Relative to Practice of Electrical Engineering states that
A. Personnel required
B. Preparations of plans, supervision of Installation, application of PEC code.
C. Practice not allowed for firms and corporations
D. Prohibitions in the Practice of Electrical Engineering

52. Which one is the portion of an instruction cycle where the Instruction is sent from memory to the Instruction register?
A. LOAD C. EXECUTE
B. ACCIMULATE D. FETCH

53. What is the weight percent of mathematics subjects?
A. 40 C. 45
B. 30 D. 25

54. Objects that emit their own light are called___bodles
A. Solar C. Photovoltaic
B. Luminated D. Illuminated

55. What is the formula for a straight-line depredation rate?
A. 100 percent-percent net salvage value / Estimated service life
B. 100 percent-percent salvage value / estimated service life
C. Percent net salvage value / estimated service value
D. average net salvage value / Estimated service life

56. This principle states that" Where the velocity of a fluid is high, the pressure is low, and where the velocity is low, the pressure is high".
A. Archimedes' Principle C. Flow Principle
B. Torricelli's Theorem D. Bernoulli's Principle

57. It is defined as a measure of the amount of negative ions in the water
A. Alkalinity C. Acidity
B. Turbidity D. Solubility

58. The material with the highest best conductivities are the
A. gases C woods
B. liquids D. metals

59. Those fund that required to make the enterprise or project aa going concern
A. accumulated amount C. working capital
B. banking D. present worth

60. RA 9136 has four sectors, which of the following is not included?
A. supply C. generation
B. distribution D. system operation

61. Which of the following is/are the disadvantage/s of operating an ungrounded electrical system?
I. Translent overvoltage cannot be controlled.
II. Power system overvoltage are not controlled.
III. Destructive arching burndowns can occur a second fault.
A. only C. II only
B. I, II, and III D. I and II

62. Which of the following Is NOT a hazardous practices?
A. Using the right tools/equipment
B. Using Improvished tools
C. Servicing equipment In motion
D. Operating without authority

63. Which of the following insulation color Identify the grounded conductor of a branch circuit?
A. Black C. Yellow
B. Green D. White

64. What Is the factor (A/P, 1%,n)?
A. Uniform series present worth factor
B. sinking fund factor
C. Uniform series compound amount factor
D. capital recovery Factor

65. Which of the following corresponds to the meaning of annuity?
A. A lump sum at the end of the year
B. A series of uniforms amounts over a period of time
C. The future worth of a present amount
D. An annual repayment of a loan

66. A rigid container is hosted by the sun. There is NO shaft work associated with the container. From the first law of thermodynamics, what is the resulting work equal to?
A. The change in Internal energy
B. zero
C. The heat transfer
D. The volume times the change in pressure

67. What percent of human body is carbon?
A. 25% C. 15%
B. 16% D. 20%

68. Which of the following is TRUE for a propeller type Impeller centrifugal pump?
A. The head developed is due to the push of the vanes and flow is almost entirely axial
B. The head developed cannot be determined by the rotation of the vanes
C. The head developed is due to the rotation of the Impeller
D. The head developed is due to the action of centrifugal force

69. Which of the following treatments will result in an Increase in the fatigue strength of the steel?
A. Shot peening C. Cold working
B. Surface roughening D. Annealing

70. Which of the following is the basic symbol used a beginning or termination in logic diagram or flowchart?
A. Square box C. Rectangular box
B. Oval symbol D. Diamond box

71. What is the transfer of heat due to motions of a fluid near the surface of an object?
A. Insulation C. Convection
B. Radiation D. Conduction

72. Which of the following do you called the brain of any computer system?
A. Arithmetic logic unit
B. Central processing unit
C. Random access memory
D. Control processing system

73. What is the term used when an American citizen is allowed to practice his profession, e. g. electrical engineering profession, here in the Philippines?
A. APEC registry C. Foreign Reciprocity
B. ACPE registry D. ASEAN Engr.

74. Under very slow deformation and high temperature, it is possible to have some plastic flow in a crystal at a shear stress lower than a critical shear stress. What is the phenomenon called?
A. Sip C. creep
B. twinning D. bending

75. Which language is used to create macros in MS Excel?
A. Visual basic C. Visual C++
B. C D. Java

76. How many times the diameter of the largest raceways shall the length of the box be in straight pulls?
A. 6 C. 8
B. 4 D. 10

77. The hardest form of carbon is
A. Coke C. diamond
B. graphite D. Charcoal

78. What is the only gas n group VIA in the periodic table?
A. Argon C. Oxygen
B. krypton D. Helium

79. Which of the following is not a control volume?
A. Insulated tank C. compressor
B. car radiator D. turbine

80. The moment of inertia of any plane figure can be expressed in units of length to the
A. First power C. Third power
B. Second power D. fourth power

81. Wirch of the following does not threaten the operation of a computer?
A. Trojan horse C. Firewall
B. Worm D. Hacking

82. Combination of laminar and turbulent flow.
A. Critical C. laminar
B. continuous D. turbulent

83. Sec. 34 of Art. IV: sundry provision Relative to practice of Electrical Engineering states that_____
A. Personnel required
B. Preparation of plans, supervision of Installation application of PEC code
C. Practice not allowed for firms and corporations
D. Prohibitions in the Practice of Electrical Engineering

84. Cavitation is the result of
A. Static pressure in a fud becoming less than fluid vapor pressure
B. Exposure of concrete to salt water
C. Hent treatment of a low carton steel
D. Improper welding technique

85. The executive officer of the Board of Electrical Engineering is_____
A. PRC Commissioner
B. Civil Service Commission
C. Chairman of the Board of E
D. IIEE President

86. Which of the following Is NOT an extensive property?
A. Kinetic Energy C. Mass
B. Momentum D. Density

87. A force causing a rise in liquid in tubes of small diameter.
A. Cohesion C. Capillarity
B. Surface tension D. Adhesion

88. The ability to convert assets to cash quickly is known as_____
A. Solvency C. Leverage
B. Liquidity D. Insolvency

89. Recognizable as suitable for the specific purpose, function, use, environment application, where described in a particular Code requirement.
A. Listed C. approved
B. labeled D. Identified

90. Which cast iron is hard and wear resistant?
A. Gray iron C. White iron
B. Ductile iron D. Malleable iron

91. Which of the following is a natural magnet?
A. Steel C. Lodestone
B. magnesia D. Soft iron

92. Month of March, is fire prevention month_____Is the electrical safety month.
A. February C. June
B. April D. May

93. Which of the following expression in incorrect?
A. The future worth of a present amount, (F/P, I, n) = $\frac{1}{(P/F, I, n)}$
B. The future worth of annuity, (F/A, I, n) $= \frac{1}{(\frac{A}{F}, i, n)}$
C. (A/F, I, n) x (P/A, I, n) = (P/F, I, n)
D. (A/F, I, n) – I = (A/P, I, n)

94. Also known as PRC Renewal Law
A. RA 8189 C. RA 7920
B. RA 8981 D. RA 184

95. Which of the following does NOT affect owner's equality?
A. Dividends paid
B. Expense to get license of start business
C. Invested capital
D. license to start business

96. What is the metal used to recover copper from a solution of copper sulphate?
A. Na C. Hg
B. Ag D. Fe

97. Atoms or lons which have the same electronic configuration?
A. Iso-lonic C. Isotope
8. Isometric D. isoelectronic

98. Kelvin-Planck's law deals with
A. Conservation of work
B. conservation of work into heat
C. Conservation of heat
D. conservation of heat into work

99. What is the factor (A/F, 1%, n)7
A. Uniform series present worth factor
B. sinking fund factor
C. Uniform series compound amount factor
D. capital recovery Factor

100. The residential value of a company's assets after all outside liabilities (shareholders excluded) have been allowed for
A Dividend C. Return
B. Equity D. Par value

SET 3

1. What contains in each cell site of cellular telephone system?
   A. repeater C. direct-link to branch
   B. touch tone processor D. control computer
2. In the periodic table of elements, which one of the following is a light metals?
   A. gold C. potassium
   B. mercury D. silver
3. What do you call a frame work composed of members joined at their ends to form a rigid structure?
   A. machine C. joist
   B. truss D. purlin
4. Is the worth of property as an operating unit
   A. scrap value C. use value
   B. value D. book value
5. The ratio of the velocity of light in vacuum to its velocity of another material is
   A. Specific Index C. Refractive index
   B. Poisson's ratio D. Modulus ratio of Elasticity
6. Hydrogen reacts with which element to form ammonia?
   A. oxygen C. nitrogen
   B. carbon D. sulfur
7. Which of the following is usually assumed to be reversible?
   A. paddle wheel C. melting an ice cube
   B. compressing a gas D. a burst membrane
8. The loss of zinc in the presence of certain corrosive media or at high the temperature is known as
   A. perzincification C. loszincification
   B. dezincification D. brazincification
9. What is produced when calcium carbide Is mixed with water?
   A. Acetylene C. propylene
   B. ethylene D. methane
10. A megabyte is exactly___bytes.
   A. 1,084,567 C. 1,048,576
   B. 999,999 D. 1,000,00
11. In RA 9136, what do you called a market that there is no choice to select a supplier of electrical energy?
   A. captive market
   B. industrial and commercial section
   C. residential section
   D. contestable market
12. The engines on a commercial jet aircraft operate on which of the basic cycles?
   A. Otto C. Carnot
   B. diesel D. Brayton
13. What is the temperature when water and vapor are in equilibrium with the atmospheric pressure?
   A. ice point C. critical point
   B. steam point D. freezing point
14. __is formed when a pair of electrons are shared between two atoms.
   A. triple bond C. covalent bond
   B. molecular orbital bond D. ionic bond
15. Only if a force on a particle is conservative:
   A. is its work zero when the particle moves exactly once around any closed path
   B. is Its work always equal to the change in the kinetic energy of the particle
   C. does it obey Newton's second law
   D. does it obey Newton's third law
16. Which of the following is an instruction that causes data to be brought from memory into an accumulator register?
   A. load C. fetch
   B. loap D. jump
17. PEP represents which one of the following?
   A. Philippine energy program
   B. Philippine energy plan
   C. Philippine electric plan
   D. Philippine energy project
18. What is the standard color for safety signages to indicate the presence of radiation hazards example In X-ray laboratory in the hospital?
   A. Green with combination of red
   B. Orange
   C. Blue
   D. Purple with combination of yellow
19. What is a coal that has been previously burned in an oxygen-poor environment?
   A. Tuyere C. Ore
   B. Diamond D. Coke
20. What is the interest acquired over the interest period is also subject to the interest rate in the next period?
   A. Quarterly Interest C. Simple Interest
   B. Compound Interest D. Annual Interest
21. Which of the following are used in designing and documenting complex processes or programs that help to visualize what Is going on and thereby help the viewer to understand a process or algorithm?
   A. Charts C. Flowcharts
   B. Diagrams D. Networks
22. Which of the following best defined the phrase "Likely to become energized?
   A. Failure of Insulation
   B. High-voltage equipment
   C. Near energized electrical equipment
   D. Equipment that Is exposed to lightning
23. Which of the following reason/s why reactions generally proceed faster a higher temperature?
   I. The molecules collide more frequently.
   II. The activation energy is less.
   A. I only C. II only
   B. Either I or II D. Both I and II
24. How tall does a wall mirror have to be in order for you to see all of yourself in the mirror? Imagine you stand a distance "d" from the mirror and your height is "h" and "L" is length of mirror required.
   A. L=1/5h C. L=1/2h
   B. L=1/4h D. L=1/3h
25. What law of nature underpins the concept of a balanced chemical equation?
   A. Gay-lussac law
   B. Faraday's law
   C. Second law of thermodynamics
   D. First law of thermodynamics
26. The shape of the bending moment diagram over the length of beam, having no external load Is always_.
   A. Cubical C. Parabolic
   B. Circular D. Linear
27. Which of the following portion of hard disk space served as RAM?
   A. Cache C. Read only memory
   B. Random memory D. Virtual memory
28. Which of the following is CORRECT regarding the sizing of the rating of the overcurrent protective device of a circuit?
   A. 125% of the non-continuous load plus 80% of the continuous load
   B. 125% of the continuous load plus 80% of the continuous load
   C. 125% of the non-continuous load plus100% of the continuous load
   D. 125% of the continuous load plus 100% of the non continuous load
29. What refers to ions that are NOT involved in the overall reactions?
   A. Guest ions C. Extra ion
   B. Spectator ions D. Special lons
30. Which of the following cast iron is heat-treated for ductility?
   A. Gray C. Ductile
   B. Malleable D. White
31. In house wiring, ground conductors must not be___.
   A. Insulated C. place in a circuit
   B. Identified D. fused
32. Which of the following describes a streamline?
   I. It is a mathematical concept
   II. It cannot be crossed by the flow
   III. It is a line of constant entropy
   A. I only C. II only
   B. I and II D. I, II, and III
33. The moment of Inertia of any plane figure can be expressed in units of length to the
   A. First power C. Third power
   B. Second power D. fourth power
34. What law asserts that energy has quality as well as quantity?
   A. First law of thermodynamics
   B. third law of thermodynamics
   C. Second law of thermodynamics
   D. zeroth law of thermodynamics
35. The original amount of money invested, borrowed, or deposited.
   A. Interest C. Funds
   B. Principal D. Stock

36. The iron that has been cooled quickly from a molten state is known as
A. Blue cast iron C. Brown cast iron
B. Pink cast iron D. White cast iron

37. After Installing a new modem and router to a computer, what is the order of turning on?
A. Computer, modem, router
B. modem, computer, router
C. Router, modem, computer
D. modem, router, computer

38. The differential of the shear equation is which of the following?
A. Load of the beam
B. bending moment of the beam
C. Tensile strength of the beam
D. slope of the elastic curve

39. A computer_____is self-replicating computer program.
A. Viruses C. Spyware
B. Worms D. Malware

40. Property of a fluid by which its own molecules are attracted is called
A. Adhesion C. Viscosity
B. Cohesion D. Compressibility

41. It state that if two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other.
A. first law of thermodynamics
B. third law of thermodynamics
C. zeroth law of thermodynamics
D. second law of thermodynamics

42. Partially protected locations under canopies, marquees, roofed open porches, and like locations, and interior locations subject to moderate degrees of moisture
A. Wet location C. Dry location
B. Damp location D. Moist location

43. What is the minimum age for a person to become a member of the Board of Electrical Engineering?
A. 30 C. 40
B. 35 D. 45

44. The property of material which allows it to be drawn into a smaller section is called
A. Plasticity C. Ductility
B. Elasticity D. Malleability

45. When a body is subjected to three mutually perpendicular stresses, of equal intensity, the ratio of direct stress to the corresponding volumetric strain is known as
A. Young's modulus C. Bulk Modulus
B. Modulus of Rigidity D. Poisson's ratio

46. What refers to a system in which there Is flow of mass across its boundaries?
A. Open system C. Isolated system
B. Closed system D. Steady flow system

47. BIOS stands for
A. Basic Input Output Society
B. Basic Input Output System
C. Base Input Output Society
D. Base Input Output System

48. A system or circuit conductor that is Intentionally grounded.
A. Grounded Conductor
B. Equipment Grounding Conductor
C. Grounding Conductor
D. Grounding Electrode Conductor

49. Fastest Internet connection for home users nowadays.
A. DSL C. FTTH
B. Dial-Up D. Broadband wireless

50. Which of the following is not a part of the requirements to take REE examination?
A. He is a citizen of the Philippines.
B. He is at least twenty-one (21) years of age
C. He is of good reputation with moral values.
D. He has not been finally convicted by the court of an offense involving moral turpitude.

51. If a refrigerator is left open in a dosed room, what will happen to the room temperature?
A. Room will get cold
B. Nothing happens
C. The room will gradually Increase in temperature
D. The room will slightly decrease in temperature

52. Which of the following is the basic equation of accounting?
A. Assets = Liability + Owner's equity
B. Liability = Assets + Owner's equity
C. Owners' equity = Assets + Liability
D. Owner's equity = Liability + Assets

53. The ozone layer exists primarily in what section of the atmosphere?
A. Troposphere C. Mesosphere
B. Stratosphere D. Ionosphere

54. The majority of the rocks in the earth's crust are:
A. Igneous C. sedimentary
B. metamorphic D. basalt

55. What element is added to copper to make it extremely hard?
A. Silicon C. sedimentary
B. aluminum D. zinc

56. Which of the following is not a ductile metal?
A. Aluminum C. Mild steel
B. Copper D. Lead

57. As a person moves closer to a plane mirror, the size of the image formed by the mirror
A. Remains the same
B. gets smaller
C. Get larger
D. increases, then decreases

58. When exposed to the atmosphere, which of the following liquids is coldest?
A. Oxygen C. hydrogen
B. nitrogen D. helium

59. A boy throws a ball with negligible air resistance. What happen to its acceleration?
A. Remains constant
B. Decelerate when the ball is going up and accelerates when going down
C. Increases
D. Decreases

60. Any device that performs signal conversion is
A. Modulator C. Keyboard
B. Modem D. Plotter

61. Air conditioning load has a demand load of____
A. 80 percent C. 125 percent
B. 100 percent D. 150 percent

62. Deformation per unit length in the direction of force is known as
A. strain C. linear strain
B. lateral strain D. unit strain

63. Which of the following diffuses most quickly?
A. Sold C. Liquid
B. Gas D. None of these

64. Which of these expressions for a one dimensional array, A, is acceptable? Assume B, H, I, J, K, and L are greater than zero.
A. A(I,K) C. A(2*L-1)
B. A(H+J) D. A(-3)

65. Which is the correct DIMENSION statement for a 9 x 9 matrix, A, and a vector, B, with nine elements?
A. DIMENSION A(9. 9),E(9)
B. DIM A(9,9),B(9)
C. DIMENSION A(9,9),B(9)
D. DIMENSION, A(9,9),B(9)

66. Find the values of X and Y after this program segment has been performed.
X=2
Y=4
Z=X
X=Y
Y=Z
A. X=2, Y=4 C. X=4, Y=2
B. X=4, Y=4 D. X=2, Y=2

67. Of the following, which is the only acceptable FORTRAN statement?
A. DIMENSION Z(TEN)
B. WRITE(6,101) (MATRIX(J,J=1,8))
C. READ(5,99) (TIME(X),X=l,10)
D. READ(5,99) (TOTAL(N),N=1,20)

68. If A and B are false, and C is true, which of the following logical expressions will be true?
A. NOT.C.AND.A C. NOT.(A.AND.B)
B. B.AND.C.ORA D. .NOT. C

69. The ability to meet debts as they become due is known as___
A. Solvency C. Insolvency
B. leverage D. Liquidity

70. Which of the following are true statements regarding user-defined functions?
I. The function is more versatile than a subroutine as it is not limited to mathematical calculations.
II. The function is defined as a variable in the main program in.
III. The function may be used repeatedly and anywhere in the program
A. I only C. I and III
B. I and II D. II and III

71. Which of the following would have to be added to the main program as a result of the variable XINCH being changed to INCH? Before the change, the function ts
FUNCTION FOOT(XINCH)
FOOT=XINCH/12. 0
RETURN
END
A. a COMMON statement
B. a CALL statement
C. a DIMENSTON specification
D. a REAL variable declaration

72. One banker's year is equivalent to____days.
A 300 C. 365
B. 360 D. 366

73. Iron is said to be abundant in nature. About how many percent of the earth's crust is iron?
A. 10 percent C. 20 percent
B. 5 percent D. 8 percent

74. The Implementing Rules and Regulation (IRR) of RA 7920 was approved under Board Resolution No. ___
A. 16 C. 22
B. 18 D. 24

75. When gases expand and mix with other gases to fill available space, it is called
A. Vaporization C. solidification
B. evaporation D. diffusion

76. In EPIRA what does OARC stands for?
A. Open Access and Retail Competitors
B. Open Access and Retailing Competitors
C. Open Access and Retail competition
D. Open Access and Retailing Competition

77. In refrigerator, the quantity 288,000 BTU/day is called
A. Ton C. Throughout
B. Vapor heat D. Freon number

78. Which of the following is not a basic component of motion of a fluid element?
A. Translation C. angular distortion
B. rotation D. twist

79. Which of the following is not a control volume?
A. Insulated tank C compressor
B. car radiator D. turbine

80. Heat can be transferred by conduction through
A. Solids only C. gases only
B. liquids only D. all of these

81. A thin mid steel wire is loaded by adding loads in equal increments till it breaks. The extensions rooted with increasing n=loads will behave as under
A. Uniform throughout
B. first increase and then decrease
C. Increase uniformly
D. increase uniformly direst and then increase rapidly

82. Which work mode is a nonequlibrium work mode?
A. Compressing a spring
B. Energizing an electrical resistor
C. Transmitting torque with a rotating shaft
D. compressing a gas in a cylinder

83. During an adiabatic, internally reversible process, what is true about the change in entropy?
A. It is always zero
B. it is always greater than zero
C. It is always less than zero
D. it is temperature dependent

84. To determine the chemical formula of a compound, *we* need to know all of the following except the
A. density of the compound
B. dements that compose the compound
C atomic weights of the element in the compound
D. percentage composition of the compound

85. The radius of gyration of a circular section of diameter d is
A. d/16 C. d
B. d/4 D. d/2

86. How many atoms are in the unit cell of a body centered cubic structure
A. 1 C. 3
B. 2 D. 4

87. What Is the first coordination number of a body centered cubic structure?
A. 4 C. 8
B. 6 D. 12

8a. What is the first coordination number of a face centered cubic structure?
A. 2 C. 8
B. 4 D. 12

69. In the design of waterways, the "hydraulic jump is sometimes used for
A. energy dissipation
B. elimination of turbulence
C. prevention of sedimentation
D. measurement of flow

90. The change in enthalpy occur at constant,
A. temperature C. volume
B. pressure D. velocity

91. The pressure exerted in a vessel by a mixture of gases is equal to the sum of the pressure that each separate gas would exert if it alone occupied the volume of the vessel.
A. Boyle's Law C. Dalton's Law
B. Charle's Law D. Avogadro's Law

92. All the heat transfer processes require a medium of energy exchange except,
A. conduction C. forced convection
B. convection D. radiation

93. What do you call the change of phase directly from vapor to solid without passing through the liquid state?
A. deposition C. condensation
B. sublimation D. vaporization

94. Steel can be strengthened lowing except
A. quenching and temperature C. work hardening
B. annealing D. grain refinement

95. A thin walled pressurized vessel consist of a right circular cylinder with flat ends. Midway between the ends the stress is greatest in what direction?
A. longitudinal
B. circumferential
C. radial
D. at an angle of 45° to the longitudinal and circumferential direction

96. Litmus paper changes from blue to red when a solution is?
A. A blue dye C. a red dye
B. an acid D. a base

97. RA 9136 has four sectors, which of the following is not Included?
A. System operation C. distribution
B. supply D. generation

98. enthalpy of an Ideal gas is a function only of
A. Internal energy
B. the product of pressure and specific volume
C. pressure
D. temperature

99. What do you call a system in which there is a flow of matter through the boundary? This system usually encloses the device that Involves mass flow, such as compressor, turbine, or nozzle.
A. Closed system C. Isolated system
B. Open system D. Al of these

100. What system in which there is no exchange of matter with the surrounding or mass does not cross its boundaries?
A. Open system C. Isolated system
B. Closed system D. Non flow system

SET 4

1. Which of the following is used to capture real and moving objective and displays in a computer
   A. scanner C. electronic recorder
   B. video player D. digital camera
2. The Electrical Engineering shall observe such a system of moral principles and values as a set of dynamic standards to guide his conduct and way of life. It is his ____ to practice his professional according to this Code of Ethics.
   A. manner C. way
   B. right D. duty
3. A language originally designed for telescope process control
   A COBOL C. ADA
   B. FORTH D. RPG
4. A process where a physical mixture of carbide and powdered metal is heated in order to solidify the powder into a single piece.
   A. Vitrification C. alloying
   B. bonding D. sintering
5. Hydraulic gradient of a pipeline represent its
   A. pressure head C. datum head
   B. velocity head D. total head
6. What refers to a document that shows proof of legal ownership of a financial security?
   A. bond C. coupon
   B. bank note D. check
7. What refers to the negotiable claim issued by a bank in lieu of term deposit
   A. time deposit C. capital gain certificate
   B. bond D. certificate of deposit
8. During a complete or partial neutralization of acids, what is the ionic compound formed?
   A. vinegar C. salt
   B. potassium D. sulfur
9. The___of a property is the amount in which a willing buyer will pay to a willing seller for the property neither one is under compulsion to buy or sell.
   A. estimate value C. market value
   B. recovered value D. assessed value
10. The___states that in a given atom, no two electrons can have exactly the same set of quantum numbers
    A. Dalton's theory C. Boyle's theory
    B. Paul Exclusion Principle D. Bohr's law
11. A crystal of solute is added to a solution, and the crystal remains undissolved. The solution is best classified as
    A. dilute C. saturated
    B. unsaturated D. supersaturated
12. The mass moment of inertia of a cylinder *about* its central axis perpendicular to a circular cross section is
    A. directly proportional to its radius
    B. independent of its radius
    C. directly proportional to its length
    D. independent of Its length
13. The pressure of gas is directly proportional to the Kelvin temperature in
    A. Boyle's low C. Boyd's law
    B. Gay-Lussac's law D. Charles's law
14. A method identifying the location of a given data record by its name.
    A. Tracking C. Signature
    B. Linker D. Hashing
15. Sec. 14 (a) (iii) of Article III Examination *and* Registration, states that prior to commending of work, the foreign professional shall secure special permit from the___
    A. Commission C. PDEA
    B. IIEE D. Bureau of Immigration
16. The Professional Regulation Commission (PRC) was created under
    A. PD 223 C. ED 223
    B. PD 59 D. E0 59
17. The ratio of Cp and Cv is called:
    A. Isothermal Index C. Hyperbolic Index
    B. Adiabatic index D. None of these
18. Which of the following is the opposite of elasticity?
    A. Rigidity C. ductility
    B. malleability D. plasticity
19. What is the area under the curve on a temperature entropy diagram?
    A. Heat C. Entropy
    B. Work D. Volume
20. Which of the following refrigerant in highly toxic and flammable?
    A. NH3 C. CO3
    B. CO2 D. R-12
21. Oxygen is converted into ozone by
    A. Great heat C. electric discharge
    B. high pressure D. all of these
22. What rules are Including an Article IV of RA 7920?
    A. Rule 1 to 10 C. Rule 21 to 27
    B. Rule 11 to 20 D. Rule 28 to 34
23. RA 7920, Rule 2(n), defining?
    A. DECS C. Board
    B. Syllabi D. CHED
24. The ideal, reversible Carnot cycle involve four basic processes. What type of processes are they?
    A. All Isothermal
    B. two adiabatic and two isentropic
    C. All adiabatic
    D. two isothermal and two isentropic
25. A KB Is how many bytes?
    A. 1000 C. 1012
    B. 1024 D. 1048
26. What is the unit of strain?
    A. Pa C. N-m
    B. unitless D. N-m/s
27. The brain of any computer system is
    A. ALU C. CPU
    B. Memory D. Control unit
28. Which of the following is an example of an integrity violations of e-commerce security?
    A. A Website is not actually operated by the entity the customer behaves it to be
    B. A merchant uses customer information In a manner not intended by the customer
    C. A customer denies that he or she is the parson who placed the order
    D. An unauthorized person intercepts an online communication and changes its content
29. What is the physical appearance of sodium chloride?
    A. Silver metal C. Yellowish Gas
    B. White crystal D. White metal
30. Pulverized fuel Is used for
    A. Better burning C. Less radiation loss
    B. more calorific value D. Medium sized units
31. Which of the following is a variable renewable source of energy?
    A. Hydro-power C. Geothermal Energy
    B. Biomass Energy D. Solar Energy
32. Which of the following is a conventional source of energy?
    A. Coal C. Wind
    B. Solar D. Tldal
33. In cold working of metals
    A. Close dimensional tolerance cannot be maintained
    B. Poor surface finish is obtained
    C. Recrystallization temperature for steel is reduced
    D. Strength and hardiness of stee is increased
34. An inventor clams to extract 100 kJ of energy from 30°CC sea water and generate 7 kJ of heat while rejecting 93 kJ to a 10°C stratum. The scheme fails because:
    A. The temperature of the sea water is too low
    B. It violates the first law
    C. It Is a perpetual-motion machine
    D. it violates the second law.
35. Steels that are used for axles, gears and similar parts requiring medium to high hardness and high strength are known as
    A. Low carbon steels
    B. Medium carbon steel
    C. High-carbon steels
    D. Very high carbon steels
36. Also known as PRC Renewal Law.
    A. RA 8189 C. RA 7920
    B. RA 8931 D. RA 184
37. The majority of the atmosphere in which weather occurs is known as the
    A. troposphere C. sedimentary
    B. metamorphic D. volcanic
38. The zone of the atmosphere in which weather occurs is known as the
    A. Ionosphere C. troposphere
    B. mesosphere D. stratosphere

39. What is another term for modulus of elasticity?
A. Bulk modulus C. constant of elongation
B. Shear strain D. Young's modulus

40. What do you call earth and stone mixed with the iron oxides?
A. Hematite C. Gangue
B. Magnetite D. Ore

41. Which light contains wavelength of all or nearly all colors?
A. White Light C. X-rays
B. Laser D. Gamma rays

42. The purpose of manganese sulfide in steel alloying is to,
A. Increase machinability
B. reduce brittleness, combined with sulphur
C. Increase hardness
D. Increase brittleness

43. What rules are Included in Article V of RA 7920?
A. RULE 11 to 20 C. RULE 21 to 27
B. RULE 28 to 34 D. RULE 1 to 10

44. In the case of a regenerative gas turbine engine, which of the following is false?
A. The generator is improving the efficiency of the engine
B. The amount of energy required during combustion s less than that required in similar gas turbine of the power is constant
C. The exit temperature of the regenerator is higher than inlet temperature of the compressor.
D. There is no heat rejected to the atmosphere in a regenerative gas turbine engine

45. When exposed to the atmosphere, which of the following liquids is coldest?
A. Oxygen C. nitrogen
B. argon D. helium

46. What is the temperature rating of a TW insulated conductor?
A. 60 deg C C. 90 deg C
B. 75 deg C D. 100 deg C

47. Electrical symbol represented by a solid line
A. Intercom wiring C. Conduit
B. Telephone wiring D. Service entrance

48. A storage system for small amounts of data is
A. Magnetic card C. Punched card
B. Magnetic tape D. Optical mark reader

49. The place where buyers and sellers come together
A. Market C. Recreation center
B. Business center D. Buy and sell section

50. Which is most common language used in web designing?
A. C C. PHP
B. C++ D. HTML

51. Device used to pull wire through the conduit.
A. Straps C Wire tongs
B. Fish tape D. Puller

52. Which of the following elements are stored in oil to prevent explosions?
A. Lithium C. ytterbium
B. actinium D. Strontium

53. For dwelling units, the demand factor for the first 3000 volt-amperes of load Is
A. 95% C. 100%
B. 85% D. 80%

54. A type of annuity where the payments are made at the start of each period, beginning from the first period
A. annuity due C. perpetuity
B. ordinary annuity D. deferred annuity

55. Which of the following refers to the diffusion of a solvent into a stronger solution in attempt to equalize the two concentration?
A. purification C. osmosis
B. electrolysis D. hydrolysis

56. What Is RES stands in EPIRA?
A. Retail Energy in Supply
B. Retail Energy in Suppliers
C. Retail Electricity Supply
D. Retail Electricity Suppliers

57. A device used to measure atmospheric pressure is called
A. manometer C. pycnometer
B. barometer D. hydrometer

58. It Is a responsible for the procurement and administration of funds with the view of archiving the objective of business.
A. capital C. finance
B. budget D. funds

59. Which Is not a standard over current protection device rating?
A. 50 C. 225
B. 1500 D. 110

60. Which of the following holds true for any given material?
A. 1/3 ≥ G/E ≥ ½ C. 1/3 > G/F ≥ 1/2
B. ½ > G/E>1/3 D. 1/2 ≥ G/E ≥ 1/3

61. Which group of keys in the keyboard will be needed to go to the bottom of a document?
A. alt +end C. ctrl + home
B. ctrl + end D. alt + end

62. What is the default setting of paper in the printers dialog box?
A. landscape C. legal
B. portrait D. A4

63. It Is anything that occupies space and has mass.
A. matter C. meter
B. water D. atom

64. How Is molarity defined?
A. the number of moles of solute 1000g of solvent
B. the number of moles of solute in 1L of solution
C. the number of gram of formula *weights* of *solute* per liter
D. the number of gram equivalent weights of solute in one *liter* of solution

65. The case with which dislocation are able to move through a crystal under stress accounts for which of the following?
A. ductility and lower yield strength
B. ductility
C. lower yield strength
D. hardness

66. It Is a particle which his no charge.
A. proton C. neutron
B. electron D. quarks

67. EPIRA is Republic Act No. ____
A. 9136 C. 8981
B. 7920 D. 7638

68. A nonconservative force:
A. violates Newton's second law
B. violates Newton's third law
C. cannot do any work
D. none of the above

69. A force on a particle is conservative if:
A. its work equals the change in the kinetic energy of the particle
B. it obeys Newton's second law
C. it obeys Newton's third law
D. Its work depends on the end points of every motion, not on the path between

70. The wound spring of a clock possesses:
A. kinetic but no potential energy
B. potential but no kinetic energy
C. both potential and kinetic energy in equal amounts
D. neither potential nor kinetic energy

71. Suppose that the fundamental dimensions are taken to be: force (F), velocity (V) and time (T). The dimensions of potential energy are then:
A. F/T C. FV/T
B. FVT D. F/T2

72. A golf ball is struck by a golf dub and falls on a green three meters above the tee. The potential energy of the Earth-ball system is greatest:
A. just before the ball is struck
B. just after the ball is struck
C. just after the ball lands on the green
D. when the ball reaches the highest point in its flight

73. The thermal energy of a system consisting of a thrown ball, Earth, and the air is most closely associated with:
A. the gravitational Interaction of Earth and the ball
B. the kinetic energy of the bell as a whole
C. motions of the Individual particles within the ball
D. motions of Individual particles within the ball and the air

74. A block slides across a rough horizontal table top. The work done by friction changes:
A. only the kinetic energy
B. only the potential energy
C. only the internal energy
D. only the kinetic and Internal energies

75. The center of mass of a uniform disk of radius R is located:
A. on the rim
B. a distance R/2 from the center
C. a distance R/3 from the center
D. at the center

76. The __when heated decompose to yield water and anhydrous salt.
A. nitrates C. hydrates
B. carbonates D. chlorates

77. What do you call the treating of an ore with aqueous reagent to dissolve component selectively
A. calcinations C roasting
B. leaching D. smelting

78. What is the temperature when water and vapor are in equilibrium with the atmospheric pressure?
A. ice point C. critical point
B. steam point D. freezing point

79. __is formed when a pair of electrons are shared between two atoms.
A. triple bond C. covalent bond
B. molecular orbital bond D. Ionic bond

80. Which of the following statements about entropy is false?
A. an irreversible process increase entropy of the universe
B. entropy of a crystal at 0 deg F is Zero
C. net entropy change in any closed cycle Is zero
D. entropy has the units and heat can pass it

81. Which of the following is NOT an intensive property?
A velocity C. pressure
B. volume D. temperature

82. Network particle allow two or more pc to communicate unto network with the same language then commercially and standard
A. OSI C. TCP/IP
B. LAN D. WAN

83. What is the reason why Styrofoam is a good insulating material?
A. it has pure white compartment
B. It contains many tiny pockets of air
C. it is very unstable and heat capacity
D. its structure is very densely compacted

84. In cold working of metals
A. dose dimensional tolerance cannot be maintained
B. poor surface finish is obtained
C. recrystallization temperature for steel is reduced
D. strength and hardness of steel is increased

85. Which of the following statements is not correct?
A. A nominal rate of 12% per annum compounded quarterly Is the same as (12%) ÷ 4 = 3% per quarter
B. One dollar compounded quarterly at 3% for n years has a worth of (1. 03)^4n dollars
C. Compounding quarterly at a nominal rate of 12% per year is equivalent to the compounding annually at a rate of 12. 55%
D. Effective rate of return choices (A) through (C) is the difference between 12. 55% and 12%

86. A device converts vertical line codes into a computer readable information.
A. Scanner C. Cash register
B. Printer D. Barcode scanner

87. What is the maximum shearing stress of a triangle with base "b" and height "h". If the base is horizontal and the applied shearing force is at a distance "Y" from the top q.
A. 4S/bh C. 4b/Sh
B. 3b/Sh D. 3S/bh

88. In molecules of the same composition, what are variations of atomic arrangements known as?
A. polymers C. Isomers
B. monomers D. crystal system

89. The Economic Order Quantity (EOQ) Is defined as the order quantity which minimizes the inventory cost per unit time. Which of the following is not an assumption of the basic EOQ model with no shortages?
A. The demand rate is uniformed and constant
B. There is positive cost on each unit Inventoried
C. The entre recorder quantity is delivered Instantaneously
D. There Is an upper bound on the quantity ordered

90. Which of the following is NOT a computer program language?
A. FORTRAN C. LOTUS
B. COBOL D. PASCAL

91. In the Reexamination of failed Subjects In the examination for Registered Electrical Engineer, an applicant shall be allowed to retake, ___times, only the subjects in which he has obtained a grade below fifty percent (50%).
A. three C. four
B. no limit D. six

92. Which of the following refers to the branch of computer science that is concerned with making computers behave and "think" like humans?
A. Intelligence Network C. Neural Network
B. Artificial Intelligence D. Robotics Science

93. What type of knot is used to protect sockets in extension cords?
A. fire C. under-writer
B. sheep shank D. dove hitch

94. Two switches on the box under one face-plate is called.
A. two-way switch C. two-gang switch
B. double pole switch D. double throw switch

95. What law asserts that energy has quality as well quantity?
A. First law of thermodynamics
B. Zeroth law of thermodynamics
C. Third law of thermodynamics
D. Second law of thermodynamics

96. What is the property of metals that reflect the light that strikes their surface, making them appear shiny?
A. Malleability C. Plasticity
B. Ductility D. Luster

97. Ceramics area mostly_____
A. carbide C. nitrides
B. oxides D. all of these

98. When using net present worth calculations to compare two projects, which of the following could invalidate the calculation?
A. difference in the magnitudes of the projects
B. evaluating over different time periods
C. mutually exclusive projects
D. nonconventional cash flows

99. How is the material usage variance defined?
A. (quantity purchased x actual price) - (quantity purchased x standard price)
B. (quantity issued - standard quantity) (standard price)
C. (actual price-standard price) (quantity used)
D. (quantity purchased - quantity used) (actual price)

100. What is wrong with this program?
DO 20 I = 1. 5
A(I, J)-5. 8
L= 5
20 Continue
A. A cannot be a real number
B. I cannot be an integer
C. There is an excess comma in the DO statement
D. The value of I cannot be changed in the DO loop.

SET 5

1. Ideal fluid has:
   A. No surface tension
   B. No surface tension, no viscosity
   C. No surface tension, no viscosity and my compressibility
   D. Surface tension, viscosity and compressibility
2. The area of the shear diagram of a beam between any two points on the beam is equal to the
   A. Change in shear between the two points
   B. Average moment between the two points
   C. Total shear beyond the two points
   D. Change in moment between the two points
3. A general form covering an assembly or assemblies of devices for the switching controlling metering of electric power.
   A. Enclosure C. Switchboard
   B. Switchgear D. Panelboard
4. In Air conditioning of aircrafts, using air as a refrigerant, the cycle used is:
   A. Reversed Joule cycle
   B. Reversed Camot cycle
   C. Reverse Brayton cycle
   D. Reverse Otto Ode
5. The second period contains how many elements?
   A. 2 C. 6
   B. 4 D. 8
6. Mercury cannot wet a glass tube. This is due to:
   A. Cohesion C. surface tension
   B. adhesion D. viscosity
7. Which of the following type of coal rank the highest
   A. anthracite C. coke
   B. bituminous D. lignite
8. How many atoms are there per unit cell of a BBC structure?
   A. 1 C. 3
   B. 2 D. 4
9. For office buildings, a general lightning load of__VA/m2 shall be used.
   A 12 C. 24
   B. 16 D. 28
10. A market situation where there is only one seller with many buyers
    A. Monopoly C. Oligopoly
    B. Monopsony D. Oligopsony
11. What are the system programs used to translate directly an assembly language to machine language called?
    A. Compiler C. Interpreter
    B. Debugger D. Assembler
12. What best defines "annuity”?
    A. A series of equal payments occurring at equal periods of time
    B. First payment does not begin until some later date.
    C. Value of the stock as stated on the stock certificate.
    D. Difference between the present value and the worth of money at some time in the future
13. In RA 7920, section 23 of article III: issuance of certificates of Registration. The professional is required to renew license every _ years upon payment of *registration* fees
    A. 1 C. 3
    B. 2 D. 4
14. It is the ratio of the ultimate stress to the allowable stress.
    A. Proportionality constant C. bulk modulus
    B. strain D. factor of safety
15. What is the entropy of a pure substance at a temperature of absolute zero?
    A. Unity C. Infinity
    B. Zero D. Undefined
16. Which of the following closely related to d'Alembert's theorem?
    A. Newton's fist low of motion
    B. Newton's third law of motion
    C. Newton's second law of motion
    D. Bernoulli's theorem
17. A photon is
    A. a particle of electromagnetic radiation
    B. a neutral proton
    C. a neutral electron
    D. a charge neuron
18. The unit of joule can be converted to which of the following?
    A. Pa*m^2 C. Pa*m^3
    B. N kg D. Pa/m^2
19. In Rigid Metal Conduit (RMC) wiring, conduits shall be supported at least every ____ mm
    A. 1500 C. 3000
    B. 2000 D. 1000
20. What prefix is equivalent to 10 raised to the power 21.
    A. Yotta C. Zepto
    B. Zeta D. Yodto
21. The term thermodynamic comes from the Greek words therme and dynamic which means,
    A. heat power C. hent energy
    B. heat transfer D. heat motion
22. It Is a method of analyzing coal which Includes moisture, ash, volatile matter and fired carbon contents
    A Proximate analysis C. Orsat analysis
    B. Ultimate analysis D. Stoichiometry
23. If is a more comprehensive coal analysis of various *elements* present in a coal sample such as carbon, *hydrogen , oxygen,* nitrogen and sulfur.
    A. Proximate analysis C. Orsat analysis
    B. Ultimate analysis D. Stoichiometry
24. Of the following statements, which is FALSE?
    A. The FORMAT statement may or may not be executed by the program.
    B. More than one STOP statement is permitted in a program.
    C. The arithmetic IF can only be used to evaluate an arithmetic expression.
    D. No FORMAT statement is required with unformatted READ or WRITE statements.
25. What is a correct declaration form for the logical variable AZ?
    A. AZ=.TRUE. C. AZ=L
    B. AZ=1 D. AZ='TRUE
26. Where is the error in the following FORTRAN program?
    670 READ L,M,N
    680 DATA 33 40 60
    690 PRINT L+M/N
    700 PRINT L*N+M
    800 END
    A. line 670 C. line 690
    B. Ine 680 D. line 700
27. In a queueing process of customers in a store, what statistical pattern will most likely describe the arrival of customers over time?
    A. the normal law of probability
    B. the Poisson distribution
    C. the uniform law of probability
    D. the exponential distribution
28. In a queueing process of customers in a store, what type of distribution most likely governs the time between consecutive arrivals of customers?
    A. a normal probability distribution
    B. an exponential distribution
    C. a uniform probability distribution
    D. a Poisson distribution
29. Identify the FALSE statement.
    A. The primary objective of PERT is to determine the probability *of* meeting specified deadlines.
    B. PERT identifies the activities that are most likely to "bottleneck" and the activities that are most likely to stay on schedule.
    C. PERT evaluates the sensitivity to changes in the program.
    D. To apply PERT, one should develop a network representation of the project plan.
30. Which of the following Is NOT a good application of network analysis?
    A. electrical engineering
    B. Information theory
    C. the study of transportation systems
    D. Inventory theory
31. The simplex method is extremely efficient In solving which of the following classic problems?
    A. the transportation problem
    B. the assignment problem
    C. the transshipment problem
    D. the allocation problem
32. Which one of the following process is also called Hyperbolic Process?
    A. Adiabatic C. Isothermal
    B. Reversible D. Isentropic

33. Section 23 of Article III of RA 7920-Issuance of Certification of Registration. The professional is required to renew license every__year/s upon payment of registration of fee.
A. Three C. Six
B. Two D. Five

34. When installing the heatsink back into a computer after replacing a failed processor, which of the following must be done during installation?
A Thermal compound must be installed between the processor and the heatsink
B. Thermal compound must be installed between the processor and the system board.
C. An Isopropyl alcohol pad must be placed between the processor and the heatsink
D. An isopropyl alcohol pad must be placed between the processor and the system board

35. When heat Is transferred by molecular collision. It Is referred to heat transfer by
A. Conduction C. radiation
B. Convection D. conduction and convection-

36. A technician has installed a new video card and the computer will now only boot into VGA mode. Which of the following may be the cause of the problem?
A. There Is an operating system Incompatibility
B. The external monitor is not compatible
C. The new card is not DVI compatible
D. The incorrect drivers are Installed

37. A good example of kinetic energy is provided by:
A. a wound dock spring
B. the raised weights of a grandfather's clock
C. a tornado
D. a gallon of gasoline

38. Which of the following statement/s about sound is/are true?
I. All sounds are produced by the vibration of bodies.
II. Sound cannot travel through a vacuum.
III. Sound can be transmitted through some material.
A. I and II C. I and III
B. I, II and III D. II and III

39. What refers to a slender member which prevents parts of a structure moving to each other under compressive force?
A. Tie C. Arch
B. Column D. Strut

40. Based on Section 18 of the R. A. 7920, the minimum age requirement for qualification of applicants for Registered Master Electricians Examination is at least___years of age.
A. Eighteen
B. Nineteen years of age
C. Twenty one years of age
D. Twenty years of age

41. What are the basic lens formulas?
A. 1/Di = 1/F + Do ; So/Si = Do/Di
B. 1/Di +1/DO = 1/F ; So/Si=Di/Do
C. 1/D = 1/f + Do ; Si/So = Do/Di
D. 1/Di + 1/DO=1/F ; S0/Si = D0/Di

42. A description of the arrangements of atoms or ions in a material is called___
A. Mixture C. Component
B. Composition D. Structure

43. Which of the following is qualified as the Chairman of the Continuing Professional Development Council for Electrical Engineering?
A. Representative from Association of Professional Organization
B. PRBEE board member
C. Representative from Academy
D. PRC Commissioner

44. Which of the following is NOT a unit of work?
A. N-m C. Dyne
B. Erg D. Kg-sq. m/sq. sec

45. A notice issued by the system operator when the contingency Reserve is less than the capacity of the largest Synchronized Generating unit or power import from a single interconnection whichever is higher.
A. Blue Alert C. Red Alert
B. Yellow Alert D. Green Alert

46. If the radius of wire stretched by a load is doubled then its Young's modulus will be
A. Doubled C. become four times
B. halved D. remain unaffected

47. In a steady, Ideal of an incompressible fluid. Total energy at any point of the fluid is always constant. This theorem is known as
A. Euler's theorem C. Reynolds theorem
B. Naylor stockers theorem D. Bernoulli's theorem

48. Heat can be transferred by conduction through
A. Solids only C. gases only
B. Liquids only D. all of these

49. Which of the following corresponding to the meaning of annuity?
A. A lump sum at the end of the year
B. An annual repayment of the loan
C. A series of uniform amounts over a period of time
D. The future worth of a present amount

50. RA 9136 is_____
A. Electric Power Installation Reform Act
B. Electric Power Industry Reform Act
C. Energy Power Isolation Reform Act
D. Energy Power Institution Reliability Act

51. The ability to convert assets to cash quickly is known as
A. Solvency C. Leverage
B. Liquidity D. Insolvency

52. Who is the current chairman of the board of EE?
A. Jaime V. Mendoza C. Gregorio T. Cayetano
B. Francis V. Mapile D. Rodolfo B. Peñalosa

53. Protection of workers from impact penetration from falling and flying objects, blows, and from limited electric shock and bums shall be provided where there is reasonable probability of exposure to such hazards.
A. Hard Hat C. Harness
B. Toe Board D. Belt

54. Aluminum alloys find use in aircraft Industry because of
A. High strength
B. Low sp Gravity
C. good corrosion resistance
D. good weldability

55. Electricity was suddenly disrupted by a lightning strike, after 5-10 seconds, the power was automatically restored by an emergency device. What was used?
A. Safety switch
B. Automatic transfer switch
C. Double throw switch
D. UPS switch

56. Which on of the following is standard temperature and pressure (STP)?
A. 0 K and one atmosphere pressure
B. 32° F and zero pressure
C. 0° F and zero pressure
D. 0°C and one atmosphere

57. The deformation of a bar per unit length in the direction of force is known as
A. Linear strain C. shear strain
B. lateral strain D. volumetric strain

58. The point that connects the saturated liquid to the saturated vapour line is called the:
A. Triple point C. superheated point
B. critical point D. compressed liquid point

59. What color is used designate caution, limited to warning against stating, use of, or movement of equipment which is under repair or being worked upon?
A. Blue C. Red
B. Orange D. Yellow

60. What is the entropy of a pure substance at a temperature of absolute zero?
A. Unity C. Infinity
B. Zero D. Undefined

61. Which Is also called stranger gas?
A. Xenon C Argon
B. Neon D. Nitrous oxide

62. An annuity where the payments are made at the beginning of each period.
A. Ordinary annuity C. annuity due
B. deferred annuity D. perpetuity

63. what are the two classification of goods and services?
A. Local and Imported
B. consumer and producer
C. Raw and finished
D. ready-made and made-to-order

64. Which of the following is known as electrical steel?
A. Silicon steel C. carbon steel
B. stainless steel D. Cast Steel

65. What Is the color of the conductor used as a neutral?
A. Orange C. white
B. yellow D. green

66. PPE for shock and arc flash protection Includes____
A. Safety glasses C. hearing protection
B. voltage rated glasses D. all of these

67. What does PPE stand for?
A. Protective Preventative Equipment
B. Personal Preventative Equipment
C. Personal Protective Equipment
D. People's Protective Equipment

68. What Is the process of putting back the lost electrons to convert lon back to a metal?
A. Oxidation C. Reduction
B. Corrosion D. Ionization

69. Iron is said to be abundant in nature. About the how many percent of the earth's crust is iron?
A. 10% C. 20%
B. 5% D. 8%

70. Which of the following is a natural magnet?
A. Steel C. Lodestone
B. magnesia D. Soft iron

71. Metalloids have a valence of__
A. 3 C. 5
B. 4 D. unity

72. Which the following refers to the parallel axis theorem for second moment of area for strength of materials?
A. Maxwell theorem C. Steiner's theorem
B. Young's theorem D. Mohr's theorem

73. What physical property of material that refers to the point at which a material liquefies on heating or solidifies on cooling?
A. Curie pint C. Melting point
B. Specific heat D. Refractive Index

74. Which of the following elements and components is reactive in its pure form?
A. CO2 C. HCl
B. He D. Na

75. Which of the following formula is CORRECT for temperature conversion?
A. TF=5/9 Tc +32 C. TF = 9/5 Tc + 32
B. TF=5/9 Tc -32 D. TF = 9/5 Tc - 32

76. What refers to the establishing interrelationships between people and things in such a way that human and material resources are effectively focused toward achieving the goal of the company?
A. Planning C. Organizing
B. Leading D. Controlling

77. Which of the following statements is FALSE?
A. Steady flows do not change with time at any point
B. The Navier-Stokes equation is the equation of motion for a viscous Newtonian fluid
C. Bernoulli's equation only holds on the same streamline
D. The Reynolds number is the ratio of the viscous force to the inertial force

78. What is device used to determine the specific gravity of the liquids?
A. Psychrometer C. Manometer
B. Hydrometer D. Venturi meter

79. What Is the heaviest subatomic particle?
A. Electron C. Atom
B. Proton D. Neutron

80. In a mixed gas such as air, the rms speed of different molecules are___
A. Proportional to molecular mass
B. Independent or molecular mass
C. Proportional to square root of molecular mass
D. Inversely proportional to square root of molecular mass

81. The ability to respond to a re-despatch performed by system operator during a trading interval or it is a generating unit that has fast capability and can synchronized with the grid o provide its declared capacity for a minimum of 8hrs.
A. spinning reserve C. power reserve
B. contingency reserve D. backup reserve

82. The color used indicate grounding conductor large than 14mnt^2 are_____and__.
I-yellow III-gray
II-white IV-any color with 3types except green
A. I,II,III C. II,III,IV
B. I,II,IV D. I,III,IV

83. Under which of the following type of loading does fatigue occur?
A. Repeated load C. Static load
B. Plane load D. High load

84. What form of software can also be called firmware?
A. Cache memory C. System BIOS
B. Operating system D. random memory

85. Which of the following treatments will result in an Increase In the fatigue strength of the steel?
A. Shot peening C. Cold working
B. Surface roughening D. Annealing

86. In a combustion chamber, fuel is burned to raise the temperature of the medium prior to production of work through an expansion process. The heat generated during the combustion reaction is the_
A. Heat of formation C. Heat of vaporization
B. Specific heat D. Heat of reaction

87. The change in enthalpy occur at constant
A pressure C. temperature
B. volume D. entropy

88. What are the van der Waals forces?
A. Primary bonds between atoms
B. Forces not presents in liquids
C. Forces between electrons and protons
D. Weak secondary bonds between atoms

89. The modulus of rigidity of a steel member is___
A. Defined as the length divided by the moment of inertia
B. Defined as the unit shear stress divided by the unit shear deformation
C. Equal to the modulus of elasticity divided by one plus Poisson's ratio
D. A function of the length and depth

90. A printed circuit board that has (at least) slots to connect cards into. Often, they also include a CPU and memory is called ___
A. Mainframe C. Motherboard
B. IDE port D. Server

91. Which of the following computer program is NOT considered open-source software?
A. Firefox C. Linux
B. Microsoft Office D. OpenOffice. org

92. Each motor compressor shall have an overload protection set at not more than____% of the motor compressor rated load current.
A. 130 C. 175
B. 115 D. 125

93. What is the device number for differential protective delay?
A. 21 C. 50
B. 51 D. 87

94. As per RA 9136 which is known as Electric Power Industry Reform Act of 2001, a large customer has a demand of at least ___MW.
A. 1 C. 10
B. 15 D. 5

95. As per RA 9136 which is known as Electric Power Industry Reform Act of 2001, a generator Is classified as large generator if its capacity is in excess of___MW.
A. 10 C. 20
B. 15 D. 25

96. As per RA 9136 which is known Electric Power Industry Reform Act of 2001 , a voltage level exceeding 1KV up 34. 5kV is classified as,
A. low voltage C. high voltage
B. medium voltage D. extra high voltage

97. SAIDI in EPERA means,
A. system average interruption duration Index
B. system average Interruption distribution Index
C. system average Interruption demand Index
D. system average interruption difference Index

98. The rating of the disconnecting means of a hermetic motor shall be__% of the rated current of the motor.
A. 115 C. 135
B. 125 D. 175

99. Fuse in a motor circuit is for_____
I-short circuit protection
II-overloaded protection
A. II only C. Both I and II
B. I only D. Either I or II

100. Which of the following thermodynamic cycles is the most efficient?
A. Carnot C. Rankine
B. Brayton D. Diesel

SET 6

1. What type of bond exist predominantly in chlorine sodium chloride crystal
   A. Ionic bond C. covalent bond
   B. metallic bond D. polar bond
2. Which of the following represented by HTTP?
   A. Huge Text Transfer Protocol
   B. Hyper Text Transfer Protocol
   C. Hyper Text Transmission Protocol
   D. High speed Text Translate Protocol
3. What Is the actual shape of the methane CH, molecule?
   A. Tetrahedral C. square planar
   B. Linear D. pyramid
4. The element that is practically insoluble in solid copper is known as,
   A. brass C. neon
   B. lead D. boron
5. Entropy is the measure of,
   A. internal energy C. randomness or disorder
   B. heat capacity D. change in enthalpy
6. What refers to the computer's use of head disk storage to simulate RAM?
   A. Volatile memory C. Data storage
   B. ROM D. Virtual memory
7. What is the term for a deadlock situation that occurs when a program references a different page for almost every Instructions and there is not even enough memory to hold most of the virtual memory?
   A paging C. trashing
   B. scanning D. handshaking
8. Which element must be introduced to remove copper from copper sulphate, CuSO4?
   A. Hg C. Fe
   B. Ag D. Na
9. Coal heating value depends largely on which of the following
   A. size if particle C. ash content
   B. moisture content D. sulfur content
10. Which of the following materials is used for making coils of standard resistances?
    A. copper C. nichrome
    B. platinum D. Manganin
11. It is the claim of anyone to ownership.
    A. proprietorship C. equity
    B. assets D. liability
12. RA 7920 Rule 2(d) defining?
    A. CHED C. Board of EE
    B. Commission D. TESDA
13. What is Section 33 of RA7920?
    A. Personnel Required
    B. Prohibition in the Practice of Electrical Engineering
    C. Field of Practice
    D. Foreign Reciprocity
14. It is the price a buyer is willing to pay a willing seller for the stock?
    A. bond value C. market value
    B. scrap value D. stock value
15. A conductor used to connect equipment or the grounded circuit of a wiring system to a grounding electrode or electrodes.
    A. Grounding conductor
    B. Grounding Electrode conductor
    C. Grounded Conductor
    D. Grounding Equipment
16. When magna rocks cool, these rods are called
    A. Granite C. Igneous rocks
    B. sedimentary rocks D. metamorphic rocks
17. The conductor used to connect the grounding electrode to the equipment grounding conductor to the grounded conductor, or to both, of the circuit at the service equipment or at the source of a separately derived system.
    A. Grounding Conductor
    B. Grounding Conductor, Equipment
    C. Grounded Conductor
    D. Ground-Fault Protection of Equipment
18. What is the functional unit of NPC created to pursue the missionary electrification function which is abbreviated as SPUG?
    A. system power utilization group
    B. small power utilization group
    C. small power utility group
    D. standard power utility group
19. A device intended for the protection of personal that functions to deenergize a circuit or portion thereof within an established period of time when a current to ground exceed some predetermined value that is less than that required to operate the overcurrent protective device of the supply circuit.
    A. Grounding conductor
    B. Grounding Conductor Equipment
    C. Grounding-Fault Circuit Interrupter
    D. Ground-Fault Protetion of Equipment
20. The maximum bending moment of a beam simply supported at both end and subject to a total load W uniformly distributed over its length L Is expressed by the formula.
    A. WL/8 C. WL/2
    B. WL2/8 D. WL/4
21. What is the form of the moment diagram of a beam supported at both ends and carrying a uniformly distributed load?
    A. Triangle C. parabola
    B. hyperbola D. rectangle
22. A byte is composed of how many bits?
    A. 2 C. 6
    B. 4 D. 8
23. An empty glass was initially place with a cube of ice. A water Was filled into a glass such that the ice at the tip of the glass. What will happen after the ice melts?
    A. Cannot determine since no density of the ice and water are given
    B. Some spilled
    C. Some spilled on the side of the glass
    D. Nothing spiled
24. A gas in the atmosphere that traps sun's UV rays.
    A. Oxygen C. Ozone
    B. Helium D. water vapor
25. Pluto travels from perihelion to aphelion. What is the work done by gravity on Pluto?
    A. Zero C. positive
    B. negative D. undetermined
26. Carnot cycle is maximum efficiency in___engine.
    A. Steam C. reversible
    B. gas D. well lubricated
27. It is the interest earned by the principal which Is added to the principal will also earn an interest for succeeding periods.
    A. Compound interest C. Discount
    B. Simple interest D. Interest rate
28. What is the term used for the situation when FM signal dominate AM signal?
    A. captured effect C. amplitude effect
    B. dominating syndrome D. frequency effect
29. Which of the following is not full duplex?
    A. telephone C. radar
    B. Lan D. telemetry
30. Really Simple Syndication (RSS) allows you to do following except
    A. edit C. receive current feeds
    B. publishing video blog D. send current feeds
31. The ability to respond to a re-despatch performed by system operator during a trading interval or it is a generating unit that has fast capability and can synchronized with the grid o provide its declared capacity for a minimum of 8hrs.
    A. spinning reserve C. power reserve
    B. contingency reserve D. backup reserve
32. What is annuity?
    A. the future worth of a present amount
    B. a series of uniform amounts over a period of time
    C. an annual repayment of a loan
    D. a lump sum at the end of the year
33. Which of the following is true regarding the minimum attractive rate of return used in judging proposed Investment?
    A. It is the same for every organization
    B. It is not relevant in engineering economy study
    C. It is larger than the Interest rate used to discount expected cash flow from investments
    D. It is frequently a policy decision made by an organization's management
34. What is PSALM in EPIRA stands for?
    A. Private Sector Assets and Liabilities Management
    B. Private Sector Assets and Liabilities Management
    C. Power Sector Assets and Liabilities Management
    D. Power Section Assets and Liabilities Management

35. Which of the following would be Identified with a control volume in thermodynamics?
A. Expansion of the gas in a cylinder after combustion
B. Compression of air in a cylinder
C. Filling a tire with air at the service station
D. The goodyear blimp during flight

36. What is negative pressure?
A. absolute pressure C. gage pressure
B. vacuum pressure D. hydrostatic pressure

37. Receptacle outlet shall be located not more than___mm above counter top of a kitchen in dwelling units
A. 600 C. 300
B. 250 D. 500

38. Is an idea refrigeration cycle, liquid leaves the condenser and is expanded in such a manner that the enthalpy of the resulting saturated mixture. This type of expansion Is known as
A. an isothermal process C. throttling process
B. a compression process D. an isochoric process

39. The deflection of beam is
A. directly proportional to the modulus of elasticity and moment of inertia
B. Inversely proportional to the modulus of elasticity and moment of inertia
C. inversely proportional to the modulus of elasticity and moment of truth
D. inversely proportional to the weight and length

40. Refers to a change that a system undergoes from one equilibrium state to another equilibrium state is
A. Process C. phase
B. path D. cycle

41. It is formed by Injecting or pouring molten metal into the mold where it will cool and take the shape of the mold.
A. ingot C. moid
B. colloid D. crystalline

42. The amount of heat needed to change solid to liquid is,
A. condensation C. solid fusion
B. cold fusion D. latent heat of fusion

43. What refers to the negotiable claim issued by a bank in lieu of term deposit?
A. time deposit C. certificate of deposit
B. bond D. capital gain certificate

44. Which of the following does NOT transfer work to or from a system?
A. An electrical resistance heater
B. A moving piston
C. The expanding membrane of a balloon
D. A membrane that bursts

45. Which of the following anchor is designed for medium and heavy weight fastening in solid materials?
A. Another damp C. Wall grip anchor
B. Toggle bolt anchor D. Stud bolt anchor

46. Electrical equipment must be grounded to ensure that dangerous voltage on metal parts resulting from a ground fault can be reduced to a safe value.
I. TRUE
II. FALSE
A. I only C. both I and II
B. II only D. Either I and II

47. Current In an electrical circuit will take__to return to the source
A. Open the circuit in the event of an overload
B. Open or close the circuit under load
C. Only the path of least *resistance*
D. Any and all paths *available*

48. Poisson's ratio is the ratio of the
A. unit stress to unit strain
B. elastic Limit to proportional limit
C. shear strain to compressive strain
D. unit lateral deformation to unit longitudinal deformation

49. As per RA 7920, member of the board shall hold office for a term of years from the date of appointment.
A. 2 C. 5
B. 3 D. 6

50. Which of the following technique must a manager use if he decides to absorb the risk in the project?
A. Use a different technology
B. Buying insurance
C. Use a different supplier
D. Create buffer in the form of management reserve or extra time to schedule.

51. A low which states that in present fluid, the pressure exerted on it anywhere gets transmitted in all directions.
A. Archimedes' Law C. Darcy's Law
B. Stoke's Law D. Pascal's Law

52. This provides detailed Information about specific hazardous material and contains some of the following such name of substance, physical, hazard, health hazard, safe handling procedures, control measures, emergency first aid procedures, etc. What we called?
A Management safety duty sheet
B. Material signages data sheet
C. Management safety data sheet
D. Material safety data sheet

53. Encryption can provide all the following features of Information system security EXCEPT__
A. Confidentiality C. Authenticity
B. Integrity D. Privacy

54. Which of the following does "NOT" affect the owner equity?
A. Invested capital
B. Expense to get license to start operation
C. Dividend paid
D. License to start the business

55. The chairman and members of the board of electrical engineering are appointed by
A. *Secretary* of department of labor and employment
B. Chairman of the professional regulation commission
C. President of the Philippines
D. Secretary of department of energy

56. Steels that are used for axles, gears and similar parts requiring medium to high hardness and high strength are known as
A. Low-carbon steels
B. High-carbon steels
C. Medium-carbon steels
D. Very high-carbon steels

57. The element that is practically insoluble in solid copper is known as
A. Lead C. Boron
B. Brass D. Neon

58. What is the form of the moment diagram of a beam supported of both ends without external load?
A Triangle C. Circular
B. Linear D. Parabola

59. What is the smallest unit if information handled by a computer?
A. Bit C. Cell
B. Byte D. Character

60. In RA 7920, what rule is the date and place of exam?
A. 9 C. 10
B. 12 D. 11

61. How many atoms are there per unit call for a face-centered cubic structure?
A. 1 C. 3
B. 2 D. 4

62. What is the most undesirable of all elements commonly found in streets?
A. Phosphorous C. Silicon
B. Manganese D. Sulfur

63. Motors must be labeled with the following nameplate rating, manufacturer's name and etc. This is
A. Manufacturing rule C. wiring design rule
B. Construction rule D. Installation rule

64. In sizing of wire this is
A. Manufacturing rule C. wiring rule
B. Construction rule D. Installation rule

65. The first law period contains how many elements?
A. 2 C. 6
B. 4 D. 8

66. Which of the following is the poorest conductor of electricity?
A. Carbon C. Steel
B. Silver D. Aluminum

67. The speed with which a liquid escape from a vessel through art orifice Is given by
A. Archimedes' Principle
B. Torricelli's Theorem
C. Bernoulli's theorem
D. flow equation

68. The most abundant dement In the universe Is
A. He C. H
B. C D. 0

69. The maximum bending moment in a simple supported beam of span L and carrying a central concentrated load "W" is
A. WL C. WL/4
B. WL/2 D. 2WL

70. Which engine is use for fighter bomber plane?
A. turboprop C. ramjet
B. turbojet D. pulsejet

71. in thermodynamics we focus our attention on which forms of energy?
A. kinetic C. potential
B. Internal D. total

72. The purpose of copper in steel alloying is to increase
A. atmosphere corrosion resistance
B. corrosion resistance
C. hardness and strength
D. deoxidation

73. The abbreviation for binary digit
A. bid C. bit
B. bin D. byte

74. A carwash area is classified as
A. Wet location C. Damp location
B. Dry location D. Moist location

75. Installations underground or In concrete slabs or masonry in direct contact with the earth.
A. Wet location C. Damp location
B. Dry location D. Moist location

76. What impurity in steel can cause "red shortness" which means the steel becomes unworkable at high temperature?
A. Sulfur C. Manganese
B. Silicon D. Phosphorus

77. A system or circuit conductor that is intentionally grounded.
A. Grounding conductor
B. Grounded conductor
C. Grounding electrode conductor
D. Grounding equipment

78. A conducting connections, whether intentional or accidental, between an electrical circuit or equipment and the earth, or to some conducting body that serves in place of the earth
A. Grounded C. Short
B. Ground D. Fault

79. Aluminum can be added to the steel to produce
A. killed steel C. live steel
B. dead steel D. first steel

80. Which of the following is an alkaline earth metal?
A. Ba C. U
B. T D. Na

81. The second law of thermodynamics can be stated as:
I-heat flows spontaneously from hot object to a cold one, but not the reverse;
II-natural processes tend to move toward a state of greater disorder or greater entropy.
A. I only C. I and II
B. II only D. Neither I or II

82. Structural steel elements subjected torsion develop
A. Tensile stress C. shearing stress
B. compressive stress D. bending stress

83. Formula of kinetic energy:
A. v^2/2m C. 2mv2
B. mv^2/2 D. mgh

84. The three-moment equation may be used to analyze
A. a continuous beam
B. a step tapered column
C. a beam loaded at the third points
D. a three element composite beam

85. ALU stands for
A. Arithmetic Logic Unit C. Arithmetic Logical Unit
B. Arithmetic Logic Users D. Arithmetic logical User

86. Dual combustion cycle is used in:
A. Petrol Engine C. Steam Engine
B. Diesel Engine D. Gas Engine

87. In a throttling process
A. There Is no work done
B. there is no change in enthalpy
C. There is no heat transfer
D. all of these

88. The formula of EER in relation to COP is,
A. EER=COPx3. 42 C. EER=COPx4. 32
B. EER=COPx342 D. EER=COPx3. 24

89. In dwelling units, the horizontal distance between outlets in a hallway is___mm.
A. 1800 C. 3000
B. 1000 D. 1500Z

90. What is the first coordination number of a face centered cubic structure?
A. 2 C. 8
B. 4 D. 12

91. Which of the following would most likely required steel containing 0. 6% carbon that has been spheroidized, cold drawn and slightly tempered?
A. a bridge C a cutting tool
B. a water pipe D. a ball bearing

92. What Is meant by controlled volume?
A. Isolated system C. fixed region in space
B. closed system D. reversible process

93. It is a common device for measuring the absolute pressure
A. barometer C. dynamometer
B. manometer D. piezometer

94. It can be used to indicate small pressure difference
A manometer C. piezometer
B. barometer D. dynamometer

95. It is a turbine consist of a series of blades mounted around wheels
A. Frands C. Impulse turbine
B. Kaplan D. Propeller turbine

96. At equal volume, the same temperature and pressure conditions, the gases contain the same number of molecules
A. Boyle's law C. Dalton's law
B. Charles law D. Avogadro's law

97. It is the energy stored in the body
A. enthalpy C. internal energy
B. heat D. entropy

98. As per RA 9136 which is known as Electric Power Industry Reform Act of 2001., SAIFI means,
A. System Average Interruption Form Index
B. System Average Interruption Frequency Index
C. System Allowable Interruption Frequency Index
D. System Allowance Interruption Form Index

99. What is the type of communication that only one of the two parties can transmit at a time.
A. full duplex C. simplex
B. half duplex D. bicomm

100. General lighting load of an office as PEC requirement is VA/sq. m
A. 24 C. 16
B. 28 D. 18

SET 7

1. What is the most common reinforcement for polymer composites?
   A. boron C. graphite
   B. ceramic D. glass fiber
2. Which of the following situations has a conventional cash flow so that an Internal rate of return can be safety calculated and used?
   A. You purchase a house and pay the bank in monthly Installments
   B. You lease a car and pay by the month.
   C. Your company undertakes a mining project in which the land must be reclaimed at the end of the project.
   D. You Invest in a safe dividend stock and receive dividends each year.
3. Which of the following are NOT units of pressure?
   A. kPa C. bars
   B. N/cm2 D. kg/m2
4. The process with constant pressure is called,
   A. Isobaric C. isovolumic
   B. isothermal D. isentropic
5. The three moment equation may be used to analyze
   A. beam loaded at the third point
   B. a three element composite beam
   C. continuous beam
   D. s step tapered column
6. The commercially used pressure measuring device is,
   A. Bourdon pressure gage C. Mercury manometer
   B. Manometer D. Piezometer
7. The effective of safe means of releasing water in reservoir is by the use of__.
   A. aqueduct C. canal
   B. weir D. spillway
8. What is the default setting of paper in the printers dialog box?
   A. Landscape C. Legal
   B. Portrait D. A4
9. NGCP warns____alert if tropical disturbance is expected to make a landfall within 24 hours.
   A. Yellow C. red
   B. blue D. weather disturbance
10. What is work-in-process classified as?
    A. An asset C. an expense
    B. a liability D. owner's equity
11. What is the area under the curve on a temperature-entropy diagram?
    A. Heat C. Entropy
    B. Work D. Volume
12. Which of the following refrigerant is highly toxic and flammable?
    A. MH3 C. CO3
    B. CO2 D. R-12
13. The ratio of Co and C, is called:
    A. Isothermal Index C. Hyperbolic Index
    B. Adiabatic Index D. None of these
14. The place where buyers and sellers come together.
    A. Market C. Recreation Center
    B. Business D. Buy and sell section
15. The linear portion of the stress-strain diagram of steel is known as the
    A. Modulus of elongation C. irreversible range
    B. Plastic range D. elastic range
16. The maximum bending moment in a cantilever of span L carrying a U. D. L of Intensity w per unit length is
    A. wL^2 C. 2wL^2
    B. wL^2/2 D. wL^2/8
17. the process of mixing clay, water and other Ingredients to make brick is known as
    A. kneading C. pugging
    B, molding D. drying
18. The process of detecting, locating and correcting logic errors in a program.
    A. Compiling C. detecting
    B. reading D. debugging
19. Installations underground or in concrete slabs or masonry in direct contact with the earth.
    A. Wet location C. Damp location
    B. Dry location D. Moist location
20. MS-WORD is a type of?
    A. System software C. application software
    B. operating system D. system architecture
21. What Is the mast common reinforcement for polymer composites?
    A Boron C. Graphite
    B. Ceramic D. Glass fiber
22. A process in which pressure Is constant.
    A. Isobaric C. Isometric
    B. Isochoric D. Isothermal
23. What is CD ROM?
    A. Compact Disk Read Orly Memory
    B. Compact Date Read Only Memory
    C. Compatible Disk Read Only Memory
    D. Compatible Date Read Only Memory
24. What is the weight percentage of electrical engineering professional subjects?
    A. 40 C. 45
    B. 35 D. 25
25. In an Ideal gas, both pressure and temperature Increase to thrice the original pressure and temperature respectively. If the original volume is Vo1 then the new volume is
    A. 2Vo C. Vo/2
    B. Va D. 4Vo
26. For churches, a general lighting load of_VA/㎡ shall be used.
    A. 12 C. 8
    B. 4 D. 2
27. No kinetic energy is possessed by:
    A. a shooting star
    B. a rotating propeller on a moving airplane
    C. a pendulum at the bottom of Its swing
    D. an elevator standing at the fifth floor
28. Which one of the following five quantities CANNOT be used as a unit of potential energy?
    A. watt second C. joule
    B. gram cm/s^2 D. kg*m^2/s^2
29. Which of the following is the definition of a joule?
    A. a unit of power C. a kg*m/s2
    B. a Nm D. a rate of change of energy
30. What is the amount of heat needed to tum 1 kg of the substance at its melting point from the solid to the liquid state?
    A. Heat of fusion C. Heat of condensation
    B. Heat of vaporization D. Heat of fission
31. The rocks generally used for roofing ls
    A. Granite C. slate
    B. basalt D. trap
32. Which these statements is TRUE for twinning?
    A. It is the most significant form plastic deformation
    B. It cannot be cause by impact of thermal structures.
    C. It occurs at lower shear stresses than slip.
    D. It frequently occurs in hexagonal dosed pack structures.
33. The PEC part 1 2017 edition Is intended as a design specification or an instruction manual to qualified person.
    I. TRUE
    II. FALSE
    A. Both I or II C. II only
    B. Either I or II D. I only
34. An open system of first law of Thermodynamics should be utilized for all the following EXCEPT__
    A. A turbine
    B. A piston-cylinder device with no inlet/exhaust valves.
    C. A compressor
    D. A nozzle
35. What does USB stands for?
    A. Universal Serial Base C. Universal Serial Bus
    B. Universal Series Base D. Universal Series Bus
36. Which of the following logic family is fastest of all?
    A. TTL C. RTL
    B. DCTL D. ECL
37. At what point is the applied force minimal in opening a hinge door?
    A. anywhere C. farthest the hinge
    B. at the center D. near the hinge
38. The slope on the road surface generally provided in the curves is known as
    A. angle of friction C. angle of banking
    B. angle of repose D. none of these
39. Which device of computer operation dispenses with the use of keyboard?
    A. Joystick C. Mouse
    B. Light pen D. Touch

40. Sec. 23 of Art III: Issuance of Certificates of Registration. The professional is required to renew license every___years upon payment of registration fees,
A. 1 C. 3
B. 2 D. 6

41. Capitalized cost of any property is equal to the
A. annual cost
B. first cost + cost of perpetual maintenance
C. first cost + Interest of the first cost
D. first cost + salvage value

42. The moment produced by two equal and opposite and collinear forces is known as a
A. resultant C. torque
B. couple D. equilibrant

43. Most of the weather phenomena takes place in the
A. mesosphere C. Ionosphere
B. stratosphere D. troposphere

44. Simple stress is often called
A. direct stress C. total stress
B. vertical stress D. all of these

45. Which function key is used to check spellings?
A. F2 C. F7
B. F5 D. F11

46. Each of the following may result to overcurrent EXCEPT
A. Overload C. Short Circuit
B. Ground Fault D. Open Circuit

47. Most wires used in residential house wiring are usually Insulated by which of the following?
A. Asbestos C. Thermoplastic
B. Cotton D. Vamished cambric

48. When a source of sound of a listener are both in motion relative to the air, the pitch of the sound heard by the listener is not the same as when source and listener are at rest. This phenomenon refers to?
A. carrier wave C. delay effect
B. reverberation effect D. Doppler effect

49. What is the reason why incidents are investigated?
A. To prevent or control a future recurrence on a similar incident
B. To determine the causes of incident
C. To gather evidence of the incident
D. To determine who is at fault in the incident

50. Which of the following is NOT a unit of work?
A. N-m C. Dyne
B. Erg D. Kg-sq. m/sq. sec

51. Which of the following is/are hazardous practice/s?
A. Using equipment improperly
B. Failure to warn or secure
C. All of these
D. Horseplay

52. Which of the following is a volatile memory?
A RAM C. PROM
B. ROM D. EPROM

53. Article V Code of Ethics states that:
This Code shall take effect upon approval by the _____ and after fifteen (15) days following its publication in the Official Gazette.
A. Commission C. DEE
B. PRC D. CHED

54. What is the temperature rating of THHN Insulation?
A. 60 deg C C. 85 deg C
B. 90 deg C D. 75 deg C

55. A type of annuity where the payments are made at the start of each period, beginning from the first period.
A. Ordinary annuity C. Annuity due
B. Deferred annuity D. Perpetuity

56. Which of these FORTRAN format specifications reads the input -567610 as-5676. 10?
A. F6. 2 C. F7. 3
B. F7. 2 D. F0. 2

57. The Philippine Electrical Code Part 1 2009 was based on the National Electrical Code 2005 while the Philippine Electrical Code Part 1 2017 Edition was based on the National Electrical Code___
A. 2017 C. 2003
B. 2011 D. 2014

58. The total load on overcurrent device located in a panel board shall NOT exceed a certain percentage of Its rating. What is the percentage?
A. 120% C. 100%
B. 90% D. 80%

59. Materials in this group are composed of one or more metallic elements such as iron, aluminum, copper, titanium, gold, and nickel. What Is this group called?
A Metals C. Solids
B. Metalloid D. Crystals

60. What is the UPS stand for?
A. Uninterrupting Power Supply
B. Uninterruptible Power Supply
C. United Power Supply
D. Uninterruptible Portable Supply

61. The most effective alloying Ingredient in Increasing hardness Is known as
A. Boron C. Silver
B. Silicon D. Sodium

62. Recognizable as suitable for the specific purpose, function, use environment application, where described in a particular Code requirement
A. Listed C. approved
B. labeled D. Identified

63. What Is the MINIMUM size in mm required for stainless stee rod as grounding electrode?
A. 12 C. 12
B. 16 D. 10

64. Which of the following is not fundamental rule in PEC?
A. Construction Rule C. Installation Rue
B. Wiring rule D. Manufacturing Rule

65. A___is a device that merges several low speed transmission into high speed transmission.
A. server C. multiplexer
B. voltage regulator D. mainframe

66. All of the following groups of plastics are thermoplastic, except:
A. phenolics, melamine and epoxy
B. polyvinylchloride (PVC) and polyvinyl actate
C. polyethylene, polyprophylene and polystyrene
D. tetraflouroethylene (Teflon) and other fluorocarbons

67. The techniques of finding the intensity and phase of light at any point, only when a portion of wave front is expected
A. quantum theory C. Huygen's theory
B. Brewster's theory D. polarization theory

68. The pitch of a note is dependent upon
A. velocity of sound C. frequency
B. amplitude D. wavelength of sound

69. Sec. 14 (a) (ii) of Art. III: Examination and Registration, states that prior to commending of work, the foreign professional secure permit from the
A. commission C. POEA
B. IIE D. Bureau of Immigration

70. RA 7920 Rule 2(d) defining?
A. CHED C. Board of EE
B. Commission D. TESDA

71. Electrical symbol represented by a broken line
A. open wiring C. underground wiring
B. Circuit homeroom D. emergency wiring

72. Which of the following engines is normally the least efficient?
A. gasoline engine C. gas turbine
B. diesel engine D. Carnot engine

73. It requires Q of heat to make a cup of coffee. How much energy is needed to make 3 cups of coffee?
A. 3Q C. Q
B. 2Q D. 4Q

74. Which of the following statements about work for a quasiequilibrium process is incorrect?
A. The differential of work is Inexact
B. Work is a path function
C. Work is the area under a P-T diagram
D. Work is always zero for a constant-volume process

75. what is BASIC stands for?
A. Basic All-purpose Symbolic Instruction Code
B. Basic All- Purpose System Instruction Code
C. Beginner's All-purpose Symbolic Instruction Code
D. Beginner's All-purpose System Instruction Code

76. What does PRBEE stands for?
A. Philippine Regulating Board of Electrical Engineering
B. Professional Regulatory Board of electrical Engineering
C. Philippine Regulatory Board of Electrical Engineering
D. Professional Regulating Board of Electrical Engineering

77. What is the common input device?
A. Speaker C. mouse
B. keyboard D. LCD

78. An Isentropic process:
A. Is adiabatic and reversible
B. Is adiabatic but may not be reversible
C. Is reversible but may not be adiabatic
D. is always reversible

79. Which are NOT included in duties and responsibilities of PRBEE?
A. Supervise and regulate the practice of electrical engineering in the Philippines
B. Issue special permits to individual foreign electrical engineers and electricians for specific projects
C. All of these
D. Perform such other functions and duties as may be necessary to Implement effectively of this Act

80. Yellow and Black sings mean:
A. You must do something
B. hazard or danger
C. You must not do something
D. Safe conditions

81. The overall energy program formulated and updated yearly by the DOE and submitted to Congress pursuant to R. A 7638.
A. Power Development Plan (PDP)
B. Philippine Electrical Code (PEC)
C. Philippine Energy Plan (PEP)
D. Preliminary Project Planning Data (PPPD);

82. Vinegar is solution of H2O and what acid?
A. Sulfuric acid C. phosphoric acid
B. acetic acid D. nitric acid

83. What is the other term for storing data in memory?
A Reading C. memorizing
B. loading D. writing

84. The ability of the surface of material to reflect light is called
A. Creep C. luster
B. malleability D. ductility

85. The mole is a basic unit of measurement in chemistry. Which of the following is not equal to or the same as one mole of the substance indicted?
A. 22. 4 liters of nitrogen gas at STP
B. 6. 02x1023 oxygen molecules
C. 1 g of hydrogen atoms
D. 16 g of oxygen molecules

86. The process of outsmarting the search engine
A. Google Fooling
B. Search Engine Optimization
C. Rank Enhancement
D. Search Engine Hacking

87. A particle is moving with its acceleration parallel to velocity. What is the direction of the particle?
A. Circle C. Parabolic
B. Linear D. Never

88. A market condition where there are two buyers and two sellers are called____
A. Duopoly C. bilateral duopoly
B. bilateral oligopoly D. bilateral duopoly

89. What do you call the kind of wire used to bind the service drop to the service point?
A. Joint wire C. Tie wire
B. Splice wire D. None of these

90. A light scattered perpendicular to an incident beam is partly?
A. Diffracted C. monochromatic
B. polarized D. coherent

91. What device used to measure fluid velocity as well the velocity of fluid inside a pipe?
A. Barometer C. Hydrometer
B. Pitot tube D. Thermometer

92. Which element Is the best conductor of electricity?
A. Gold C. copper
B. silver D. aluminum

93. The area of the shear diagram of a beam between any *two* points on the beam is equal to the
A. Change in shear between the two paints
B. Total shear beyond the two points
C. Change in deflection between the two points. two points
D. Change in deflection between the two points

94. In flow chart, what is the symbol used in decision making?
A. Oval C. Diamond
B. Rectangle D. Square

95. Which of the following is not a fundamental rule in PEC?
A. Construction Rule
B. Wiring Rute
C. Installation Rue
D. Manufacturing Rule

96. Reversed joule cycle is also known as
A. Rankine cycle C. Brayton cycle
B. Sterling cycle D. Bell-Colemen cycle

97. Also known as PRC Renewal Law
A. R. A. 8189 C. R. A. 7920
B. R. A. 8981 D. RA 7918

98. ASCII means
A. American Standard Code for Information input
B. American Standard Code for Input Information
C. American Standard Capacity for Information Input
D. American Standard Code for Information Interchange

99. For FREE board examination, the passing general average ls 70% provided there is no grade below _____In any of the 3 subjects
A. 50% C. 65%
B. 60% D. 40%

100. Newton's first law is often referred to as which of the following?
A. The law of straight-line motion
B. The law of Inertia
C. The law of momentum
D. F=ma

SET 8

1. What is the most common type of digital IC package?
   A. DIP C. CMOS
   B. Metal type D. TTL
2. What refers to the debugging method in which the program is executed one instruction at a time and the register contents can be examined after each step?
   A. text editing C. trace
   B. syntax analyzing D. stacking
3. A sample of a substance to which a chemical formula can be assigned, whose weight is equal to its formula weight Is termed a;
   A. mole C. gram equivalent weight
   B. molecule D. atom
4. Hard water is water which contains soluble salts of which of the following;
   A. calcium C. sulfur
   B. sodium D. chlorine
5. In sizing of wire, this is:
   A. Manufacturing rule C. wiring design rule
   B. construction rule D. Installation rule
6. Galvanized steel is steel product coated with_____
   A. Carbon C. Zinc
   B. Sulfur D. Nickel
7. Which of the following materials is not viscoelastic material?
   A. Glass C. Metal
   B. Rubber D. Plastic
8. What is the study of heat and its transformation of energy?
   A. Thermodynamics C. Thermoplastic
   B. Thermophysics D. Thermochemistry
9. A property of material which it can be rolled a wire?
   A. Ductility C. Lustre
   B. Malleability D. Elasticity
10. What is the measure of the average speed with which accounts receivable are collected?
    A. Current ratio C. Acid test ratio
    B. Quick ratio D. Receivable turnover
11. What do you call the ratio of the net income to owner's equity?
    A. Return on Investment C. rate return
    B. Gross margin D. interest expense
12. What is the minimum height of the track of a roller coaster such that it can go around a loop without falling off or going off the track?
    A. $h = \frac{1}{2}r$ C. $h = r$
    B. $h = \frac{5}{2}r$ D. $h = \frac{1}{4}r$
13. Which of the following is the process that cannot be found in a Camot engine cycle?
    A. Isobaric expansion C. Adiabatic expansion
    B. Adiabatic compression D. Isothermal expansion
14. To consider as continuous load:
    A. 3 hours or more C. 5 hours or more
    B. 4 hours or more D. 6 hours or more
15. NGCP warns_____alert if tropical disturbance within 24hours.
    A. Yellow C. red
    B. blue D. weather disturbance
16. In computing, what is a taper drive?
    A. A data storage device C. a pointing device
    B. An OCR device D. a multimedia device
17. The continuous current must not exceed_____of rating of the conductor?
    A. 60 percent C. 80 percent
    B. 70 percent D. 90 percent
18. Is an act of sending e-mails or creating web pages that are designed to collect an individual's online bank, credit card or other login Information?
    A. Malware C. Cracking
    B. Hacking D. Phishing
19. The original amount of money invested, borrowed, or deposited.
    A. Interest C. Funds
    B. Principal D. Stock
20. An Electrical Engineer shall give___to the engineering work of the person to whom the credit is legality due
    A. Loyalty C. honor
    B. advices D. credit
21. Which of the following is usually a temporary connection?
    A. Bus C. Jumper
    B. Tie Wire D. Feeder
22. An establishment or a system for the production modification of electric energy is an _____
    A. Electric supply C. Substation
    B. Electric system D. Electric plant
23. Which element melts in your hand like chocolate bar?
    A. Krypton C. strontium
    B. tantalum D. cesium
24. BCD is
    A. Binary Coded Decimal C. Binary Coded Digit
    B. Bit Coded Decimal D. Bit Coded Digit
25. Which of the following is Charles's law?
    A. V = k x 1/P C. PV = mRT
    B. P1/t2 = P2/T2 D. V/T = k
26. The continuity equation is an expression of
    A. Conservation of mass
    B. conservation of momentum
    C. Conservation of energy
    D. conservation of angular momentum
27. If you place your hand underneath, blue not touching, a kettle of hot water, you mainly feel the presence of heat from
    A. Conduction C. radiation
    B. convection D. all of these
28. Vacandes of the Board set shall be filled by the__from list of recommended selected by the commissioner who was chosen form the list of nominees submitted by the integrated and accredited association for the unexpired term only.
    A. IIEE C. Chairman of the Board
    B. President D. Commissioner
29. In a transverse wave, the individual particles of the medium
    A. Move in circles
    B. Move in ellipse
    C. Move parallel to the direction of the wave's travel
    D. Move perpendicularly to the direction of the wave's travel
30. Which of the following is the other term used for enthalpy-entropy (h-s) diagram?
    A. Enthalpy diagram C. Steam s=diagram
    B. Mollier diagram D. entropy chart
31. Who was the father of internet?
    A. Charles Babbage C. Denis Riche
    B. Vint Cerf D. Martin Cooper
32. What is true about the polytropic exponent, for a perfect gas undergoing an isobaric process?
    A. N>0 C. n→a1
    B. n<0 D. n=0
33. In molecules of the same composition, what are the variations of atomic arrangements known as?
    A. Polymers C. isomers
    B. monomers D. crystal systems
34. Which of the following are NOT examples of nonvolatile memory in the computer system?
    A. Flash drives C. Optical disk
    B. RAMS D. hard drives
35. The three moment equation may be used to analyze___
    A. An axially end-load beam
    B. a continuous beam
    C. A beam loaded at the third points
    D. a step tapered column
36. What is a protective device designed to protect an individual against electric shock?
    A. Circuit Breaker C. Differential Relay
    B. AFCI D. GFCI
37. How is "molality" defined?
    A. The number of moles of solute in 1 L of solution
    B. The number of gram-equivalent wright of solute in L of solution
    C. The number of males of solute in 1000 g of *solvent*
    D. The number of gram-formula weights of solute per liter
38. When a force is called a normal force, It is
    A. The usual force expected *given* the *arrangement* of a system
    B. A force that is perpendicular to the surface of the earth at any given location
    C. A force that is always vertical
    D. A contact force perpendicular to the contact surfaces between two solid objects.
39. Which device among following is used for sending digital data over a phone line?
    A. USB C. Printer
    B. Scanner D. Modem
40. When a vapor condenses to a liquid,
    A. Its Internal energy increases
    B. Its temperature falls
    C. Its temperature rises
    D. It gives off Internal energy

41. Which of the following classes of mass memory devices used a laser beam to write and read onto a specified coated disk?
A. Non-volatile memory
B. RAM
C. ROM
D. Optical disk memory

42. Article IV, Rule Requirements for examination as a REE
A. 14 C. 16
B. 15 D. 17

43. Modern contact surface is made from what alloys?
A. Copper C. Aluminum
B. silver D. Manganin

44. For the purpose of lightning protection, a high-rise building is a building with a height over how much?
A. 23m C. 20m
B. 50m D. 15m

45. The group VII elements, called the noble or rare gases, have the following property.
A. They are rare
B. they are lighter than air
C. They do not react other elements
D. all of these

46. A gamma ray is composed of
A. high energy protons
B. high energy electrons
C. high energy neutrons
D. high energy electromagnetic radiation

47. What refers to the manner in which alphanumeric data and control characters are represented by sequences of bits?
A. controlling C. coding
B. processing D. transforming

48. Which one refers to a device that amplifies signal and passes it on?
A. repeater C. trap
B. amplifier D. modulator

49. What do you call a communication network covering a geographical area the size of a city or suburban?
A. local area network (LAN)
B. metropolitan area network (MAN)
C. wide area network (WAN)
D. multi-level network

50. A computer that is smaller and more portable than a computer notebook which is sometimes called palm top computer.
A. microcomputer
B. personal digital assistant (PDA)
C. minicomputer
D. micro PC

51. What refers to a technique where each user of a computer takes turn, under be control of the operating system, using the entire computer main memory for a length of time?
A. multitasking C. handshaking
B. time sharing D. batch processing

52. A computer Virus that infects the macros that are attached to documents and spreadsheets is called,
A. micro virus C. document virus
B. file virus D. spreadsheet virus

53. Which of the following devices merges signal?
A. oscillator C. demodulator
B. multiplexer D. demultiplexer

54. What do you call a collection of fields?
A. file C. document
B. record D. database

55. What is the another term of operating system (OS)?
A. start up program C. monitor program
B. computer program D. all of these

56. Heat Is absorbed by the refrigerant in a refrigerator when it
A. melts C condenses
B. vaporizes D. is compressed

57. Which band of light wavelength are too short to be seen by the human eye?
A. visible C. infrared
B. ultra violet D. xc-ray

58. The rate of doing work is known as
A. power C. kinetic energy
B. potential energy D. none of those

59. Which of the following has the characteristics of both metals and non metals
A. conductors C. metalloids
B. Insulators D. meteors

60. Which of the following materials is used for making coils of standard resistances?
A. copper C. nichrome
B. platinum D. Manganin

61. What Is that part of computer instruction that defines what type of operation the computer is to execute on specified data?
A. machine language C. assembly language
B. demonic D. op code

62. What is the term for a computer or software that request that request information from another computer to server?
A host C. client
B. workstation D. All of these

63. Carbon dioxide "snow" from a fire extinguisher puts out a fire because it
A. Lowers the kindling temperature of the material
B. Displace the supply of oxygen
C. Raises the kindling temperature of the material
D. Cools the material below its kindling temperature

64. During a refrigeration cycle, heat is rejected in a:
A. Evaporator C. compressor
B. condenser D. expansion valve

65. Stainless steel resists corrosion due to
A. Carbon C. chromium
B. sulfur D. vanadium

66. Absolute reference in a spreadsheet is used by Inserting a sign before the column and row. This is used for data reference in another cell.
A. Period C. semi-column
B. dollar D. hyphen

67. Earth's surface is part of the
A. Asthenosphere C. benthosphere
B. lithosphere D. troposphere

68. This refers to the energy-consuming equipment including motors, heaters, furnace, light sources and other devices which utilize electric energy. For any purpose.
A. Industrial Plant C. Processing plant
B. Utilization equipment D. Transformer

69. What rules are included in Article V of RA 7920?
A. RULE 1 to 10 C. RULE 21 to 27
B. RULE 11 to 20 D. RULE 28 to 34

70. What is the amount of heat needed to tum 1 kg of the substance at its melting point from the solid to the liquid state?
A. Heat of fusion C. Heat of condensation
B. Heat of vaporization D. Heat of fission

71. For rectangular section of breadth "b" and depth "h" the section modulus is given by:
A. z = bh3/12 C. z = bh2/12
B. z = bh3/6 D. z = bh2/6

72. The ampacity of the fuse should be 125% of the continuous load current is covered by
A. Installation design C. manufacturing design
B. Wiring design D. construction design

73. Which of these elements has the highest atomic number?
A. Titanium C. Radium
B. Uranium D. Plutonium

74. Concurrent forces are those forces whose line of action
A. meet at one point C. lie on the same line
B. meet on the same plane D. none of these

75. A couple produces
A. combined translatory and rotational motion
B. translatory motion
C. rotational motion
D. none of these

76. A conduct bend needed if the conduit end Is running through an outlet box for connection of lighting fixtures, convenience of etc.
A. Gooseneck bend C. ZComer bend
B. Saddle bend D. offset bend

77. What Is then classification of a turbine if the water moves along the vane toward to the ands of rotation of the runner?
A. Axial flow C. Parallel flow
B. Radial flow D. Mixed flow

78. Which of the following is not a method of non-destructive testing of steel casting and forgings?
A. radiography C. ultrasonic
B. magnetic particle D. chemical analysis

79. The clearance between conductors and roof where the service conductors are passing over roofs must be mm.
A. 1000 C. 2000
B. 1500 D. 2500

80. If you are appointed as a division manager, which of the following task Is MOST likely you will do?
A. determine the resources needed
B. Develop strategies and tactics
C. set a standard
D. set goals

81. What can you say about the following statement regarding centrifugal pumps? "The discharge of the centrifugal pumps that can be throttled without the building up of an excessive pressure of overloading its driving unit. "

I. True II. False

A. Either I or II C. II

B. Both I and II D. I

82. The Central Processing Unit of computer is a combination of

A. arithmetic logic and Input unit

B. control and storage

C arithmetic logic and control unit

D. control and Input unit

83. How many times the diameter of the largest raceways shall the length of the box be in straight pulls?

A. 6 C. 8

B. 4 D. 10

84. The moody diagram is used in fluid mechanics to obtain

A. the drag coefficient C. the friction factor

B. the stroudal number D. the pressure coefficient

85. Which statement about area moments of Inertia is false?

A. The area moment of inertia arises whenever the magnitude of the surface force varies linearly with distance.

B. The moment inertia of a large area is equal to the summation of the inertia of the smaller areas with in the large area.

C. The areas closest to the axis of interest are the most s ignificant.

D. area moments of inertia are always positive

86. A force causing a rise in liquid in tubes of small diameter.

A. Cohesion C. Capillarity

B. Surface tension D. Adhesion

87. Formula for calcium chloride.

A. CaCl2 C. CaCl

B. Ca2Cl D. CaCl3

88. The first period contains how many elements?

A. 2 C. 6

B. 4 D. 8

89. Which of the following is an example of an integrity violation of e-commerce security?

A. A website is not actually operated by the entity the costumer behaves it to be

B. A merchant uses customer Information in a manner not intended by the customer

C. A customer denies that he or she is the person who placed the order

D. An unauthorized person intercepts an online communication and changes its content

90. The grounded conductor shall be identified by what color/s?

I. White II. Gray

III. Green IV. Black

A. I and II C. III and IV

B. I and IV D. II and III

91. Which of the following materials is not viscoelastic material?

A. Glass C. Metal

B. Rubber D. Plastic

92. The "lead" in a lead pencil is

A. bone black C. lead oxide

B. graphite and clay D. lead peroxide

93. What term is used to describe retrieval of data from a memory

A. unloading C. reading

B. downloading D. accessing

94. What Is the risk in operating a restaurant

A. safe C. ambivalent

B. neutral D. risky

95. In electrochemistry , oxidation is a loss of

A. lon C. proton

B. electron D. anode

96. _____ is regarded as father of computer.

A. Pascal C. Charles Babbage

B. John Napier D. Abacus

97. Gross profit, sales less cost of goods sold, as a percentage of sales is called

A. ROl ratio C. gross margin

B. profit margin D. price margin

98. At what point is applied force minimal in opening a hinge door?

A. near the hinge C. at the center

B. farthest the hinge D. any where

99. What factor must be considered by the investor before he makes investment?

A. return of investment C. return of capital

B. hurdle rate D. rate of return

100. The uncovered depredation results due to poor estimate as the life of equipment.

A. sunk cost C. in-place value

B. economic life D. annuity

SET 9

1. Who Is the ERC Chairperson?
   A. Atty. Monalisa C. Dimalanta
   B. Atty. Alexis Lumbatan
   C. Atty. Catherine Maceda
   D. Atty. Marko Fuentes
2. Who is the Chairman of PRC?
   A. Charito Zamora C. Erwin Enad
   B. Jose Queto D. Jose Rizal
3. Where does PRC belong before?
   A. DENR C. CHED
   B. DOLE D. DEP Ed
4. When did RA 10912 or CPD law was approved?
   A. June 21, 2016 C. July 21,2016
   B. July 21,2015 D. June 21, 2015
5. When did RA 7920 was approved?
   A. February 24,1994 C. February 24,1995
   B. March 24, 1994 D. March 24, 1995
6. When did RA 9136 was approved?
   A. June 8, 2001 C. July 8,2001
   B. June 8,2000 D. July 8,2000
7. The rating of a branch circuit is determined by the____
   A. the bad supplied C. overcurrent device
   B. size by the conductor D. all of these
8. What is the maximum number of overcurrent devices for a lighting and appliance load that can be installed in a single cabinet?
   A. 36 C. 21
   B. 42 D. 43
9. The motor compressor shall be protected by a separate overload relay to trip at not more than_____percent of the motor compressor load current
   A. 115 C. 130
   B. 125 D. 115
10. A fuse or inverse time current breaker rated at not more than percent of the motor compressor rated load current can protect the motor against overload.
    A. 115 C. 130
    B. 125 D. 140
11. In the straight pulls the length of the box shall not be less than times the trade diameter of the largest raceway.
    A. 4 C. 8
    B. 6 D. 12
12. Which of the following shall be provided where necessary to electrical continuity?
    A grounding C. jumpers
    B. bonding D. shunts
13. The_____shall have the responsibility of implementing the provision of the Philippine Electrical Code
    A. Board of electrical engineering
    B. Department of Public Works
    C. Office of the Building Official/EE
    D. IIEE
14. Class I locations that are hazardous because of____
    A. presence of combustible dust
    B. over 2 meters depth of water
    C. Flammable gasses or vapors are or maybe locked in air
    D. presence of easily ignitable fibers
15. Class II locations that are hazardous because of____
    A. presence of combustible dust
    B. over 2 meters depth of water
    C. Flammable gasses or vapors are or maybe locked in air
    D. presence of easily Ignitable fibers
16. The Implementing Rules and Regulations (IRR) of RA 7920 was approved under Board Resolution No. series of 1995.
    A. 16 C. 22
    B. 18 D. 24
17. The first metal used by man was
    A. Iron C. Gold
    B. Copper D. Bronze
18. In RA 9136, what Is CREM stands for?
    A. competitive retail electricity market
    B. competitor retail electricity market
    C. competitive retail energy market
    D. competitor retail energy market
19. Article II of Code of Ethics for Electrical Engineers contains:
    A. SEC. 1 to 8 C. SEC 1 to 9
    B. SEC. 1 to 6 D. SEC 1 to 7
20. Which wave requires a medium for transmission?
    A. light C. radio
    B. Infrared D. sound
21. The iron that has been cooled quickly from a molten state is known a3
    A. Blue cast Iron C. Brown cast Iron
    B. Pink cast iron D. White cast Iron
22. What does the prefix tera refers to?
    A. 1012 C. 1012
    B. 105 D. 10
23. What is PEMC In EPIRA stands for?
    A. Philippine Electricity Market Commission
    B. Philippine Electricity Market Company
    C. Philippine Electricity Market Committee
    D. Philippine Electricity Market Corporation
24. Which is scalar quantity?
    A. force C. energy
    B. displacement D. velocity
25. RA 7920 contains how many sections all in all?
    A. 40 C. 43
    B. 41 D. 45
26. The most Important element that controls the physical properties of steel is,
    A. silicon C. manganese
    B. tungsten D. carbon
27. Accounting book in which are recorded the credits and debits of commercial transaction
    A. ledger C. balance
    B. journal D. clerk's book
28. ____is a process of using a microcomputer, mouse, scanner, laser printer that combines highly digital text and Image to produce a highly quality output
    A. CAM C. CAD
    B. Desktop Publishing D. CAE
29. RISC means?
    A. Reduce Instruction Set Computing
    B. Reduce Instruction Set Computer
    C. Reduce Instruction Set Component
    D. Reduce Instruction Set Capacity
30. Salt, water, rubber, iron rust are called
    A. mix atom C. element
    B. chemical compound D. molecule
31. Is the worth of the property which is equal to the original cost less the amount which has been charge to deprecation
    A. scrap value C. fair value
    B. book value D. use value
32. The particle inside the nucleus is called
    A. nucleons C. nucleic
    B. nuclei D. nuclide
33. Under the Penalty Clause, the fine shall be
    A. not more than 60,000 C. not more than 20,000
    B. not more than 50,000 D. not more than 10,000
34. What is the maximum operating temperature of type UF conductor?
    A. 75 deg C C. 60 deg C
    B. 90 deg C D. 110 deg C
35. Class III locations that are hazardous because of____
    A. presence of combustible dust
    B. over 2 meters depth of water
    C. Flammable gasses or vapors are or maybe locked in air
    D. presence of easily ignitable fibers
36. The phenomenon of melting under pressure and freezing again when the pressure Is reduces is known as
    A. sublimation C. deposition
    B. condensation D. regelation
37. If a fluid "fails" In a piping system, the "failure" is related to which of the following properties?
    A. sulfur tension C. viscosity
    B. bulk modulus D. vapor pressure
38. The element that provides the electrical power to the components in the computer box is
    A. power button C power supply
    B. power cord D. ups
39. When gases expand and mix with others gases to fill available space, it Is called
    A. vaporization C. solidification
    B. evaporation D. diffusion
40. FTP means,
    A. file transfer protocol
    B. file transfer port
    C. file transmission port
    D. file termination protocol

41. The group VIII elements, called the noble or rare gases, have the following properties
A. they do not react with other elements
B. they are rare
C. they are lighter than air
D. they remain gaseous down to just few degrees absolute zero

42. How many electrons are there in the outmost octate of a chlorine (Cl) atom?
A. 7 C. 3
B. 1 D. 8

43. What is the actual shape of the methane CH, molecule?
A. Tetrahedral C. square planar
B. linear D. pyramid

44. Reactions generally proceed faster at higher temperature because of which of the following
A. the activations energy is less
B. the molecule are less energetic
C. the bonds are easier to break
D. the molecules collides more frequently

45. In organic chemistry, which compound families are associated with the following bonds?
1. C-C 2. C=C 3. C=C
A. 1: alkane, 2: alkene, 3: alkyne
B. 1: alkene, 2: alkyne, 3: alkane
C. 1: alkyne, 2: alkane, 3: alkene
D. 1: alkane, 2: alkyne, 3: alkene

46. What are valence electrons?
A. the outer shell electrons
B. electrons with positive charge
C. the electrons of complete quantum shell
D. the K-quantum shell electron

47. What is the strong bond between hydrogen atoms know as?
A. covalent bond C. metallic bond
B. ionic bond D. Ionic and metallic bond

48. What are the Van dear Waals forces?
A. weak secondary bonds between atoms
B. primary bond between atoms
C. forces between electrons and protons
D. force present only a gases

49. The only base unit in SI with prefix kilo is
A. kilometre C. kilojoule
B. kilogram D. kilopascal

50. A device that measures temperature by the electromotive force called thermocouple was discovered by
A. Galileo C. Celsius
B. Fahrenheit D. Seebeck

51. An anode is,
A. a pentode with a ground plate
B. the end point of direct network
C. an electrode at which oxidation occur
D. an electrode at which reduction

52. What do Impact test determine?
A. hardness C. yield strength
B. toughness D. creep strength

53. What Is the packing factor of a hexagonal closed pack (HCP) lattice?
A. 0. 52 C. 0. 68
B. 0. 60 D. 0. 74

54. Prony brake is used for testing brake horse power of
A. small engine
B. large engine
C. high speed engine
D. engine having small flywheel

55. A chemical reactions accompanied by light and heat
A. radiation C. fusion
B. combustion D. fission

56. FLAG means,
A. fiber link around the globe
B. four mode link around the globe
C. fiber optic link around the globe
D. four link around the globe

57. Any equipment which produces, modifies, regulates, or controls the supply of electric energy is the__
A. Electric supply equipment C. Transformer
B. Substation D. Generator set

58. DSL means,
A. digital subscriber line
B. digital subscription link
C. digital subscription line
D. digital subscriber link

59. PDF means,
A. portable document format
B. portable document file
C. portable document form
D. portable data file

60. An establishment or a system for the production and modification of electric energy is an
A. substation C. electric plant
B. electrical system D. electric supply

61. Article II, Section 5 - Code of Ethics states that. An Electrical Engineer shall honestly advice his clients or employer if specialize work is called for to engage the _____ of competent experts and specialist, and once engaged be shall cooperate with them interest of his client or employer.
A. Information C. experience
B. cooperation D. services

62. Files of data and programs are located in many places of your hard disk and other secondary storage devices. What keeps track where they are?
A. references C. operating system
B. supervisor D. cues

63. It refers to the fractional change in the volume of a fluid per unit change in pressure at constant temperature process.
A. bulk modulus C. capillarity
B. proportional limit D. compressibility

64. How are element numbered 90 to 103 In the periodic table called?
A. alkali C. earth metals
B. actinons D. transition element

65. _____are compound of carton in hydrogen containing only carbon-carbon single bonds.
A. alkynes C alkanes
B. alkenes D. alkali

66. Relationship between temperature and pressure at constant volume
A. Henry's law C. Gay Isaac's law
B. Charle's law D. Boyle's law

67. Which of the following metallic element has meaning heavy stone"?
A. titanium C. cobalt
B. iron D. tungsten

68. When an object exerts a force a force, the second object exerts on the first a force of the same magnitude but opposite direction. This statement is in accordance to
A. Newton's First Law of Motion
B. Newton's Second Law of Motion
C. Newton's Third Law of Motion
D. All of these

69. A graphical device used to convert true table to its corresponding logic circuit in a simple, orderly process
A. state table C. Karnaugh map
B. state diagram D. truth table

70. The product of force and time during which its acts is known as
A. momentum C. impulse
B. impact D. work

71. Service conductor made up in the form of cable.
A. Service cable C. Service Drop
B. Service Raceway D. Service Lateral

72. For hospitals, a general lighting load of ___ VA/m² shall be used.
A. 12 C. 8
B. 24 D. 16

73. The element that was discovered between calcium and titanium is___
A. Silicon C. Molybdenum
B. Scandium D. Potassium

74. Which gate is formed by inverting the output of the AND gate?
A. NAND gate C. NOR gate
B. OR gate D. None of these

75. _____ is an operating system for multiple users, with built-in networking capability to run multiple task at one time a version that can run on all kinds of computers
A. UNIX C. MS DOS
B. windows D. BASIC

76. What do you call the hearting of an ore to bring about its decomposition and the elimination of a volatile product?
A. leaching C. roasting
B. calcinations D. smelting

77. ___is a treatment in which the steel is heated over the critical temperature and allowed to cool in still air.
A. air blasting C. cool storage
B. normalizing D. water cooling

78. Battery acid is the common name for
A. formic acid C. hydrochloric add
B. nitric add D. sulphuric acid

79. Car A has a mass of 1000 kg and a speed of 60 km/h and car B has a mass of 2000 kg and a speed of 30km/h. The kinetic Inner of a car A is.
A. half that of car B C. twice that of car B
B. equal to that of car B D. four time of that car B

80. A written acknowledgement by a canier that he has received the goods for shipment.
A. crate bill C. ticket
B. bill of landing D. contract to ship

81. It Is a process of mixing water with air by spraying water or diffusing ai through it
A. aeration C. Infiltration
B. coagulation D. evaporation

82. A device used to keep moisture from passing through the system is called
A. trap C. humidifier
B. aerator D. dehydrator

83. What is the article V of Code of ethics for Electrical Engineers?
A. Professional Life
B. Penal Provisions
C. Effectivity
D. Relation with Client and Employer

84. What equilibrium state exist when a body remains in Its displaced state after being moved from its original equilibrium position?
A. neutral C. static
B. stable D. dynamic

85. Thermodynamic cycle used in a thermal power plant is,
A. Ericson C. Joule
B. Brayton D. Rankine

86. It is the energy associated with the kinetic random motion of large number of molecules
A. internal energy. C. heat
B. kinetic energy D. enthalpy

87. The true pressure measured above a perfect vacuum
A. absolute pressure C. atmospheric pressure
B. gage pressure D. vacuum pressure

88. The total entropy of pure substance approaches as the absolute thermodynamic temperature approaches zero is what law of thermodynamics?
A. Zeroth law C. Second law
B. First law D. Third law

89. Work or energy can be a function of all the following except?
A. force and distance C. power and time
B. torque and angular rotation D. force and time

90. In an electrochemistry, oxidation is a loss of
A. Ion C. proton
B. electron D. anode

91. What particles are formed in the nucleus of an atom?
A. protons and electron C. neutrons and electrons
B. protons and neutrons D. neutrons and positrons

92. An objective that has momentum must also have
A. acceleration C. kinetic energy
B. impulse D. potential energy

93. If a distribution is parallel into the direction of travel of a wave, the wave is classified as
A. longitudinal C. transverse
B. electromagnetic D. torsional

94. What element Improves the strength and fatigue properties of copper?
A. berkelium C. barium
B. americium D. beryllium

95. Share of participation is called
A. stock C. corporation
B. partnership D. franchise

96. A measure of how it emits radiant energy in comparison with a black body at the same temperature
A. absorptivity C emissivity
B. reflectivity D. conductivity

97. In a manometer the pressure is measured by balancing it by?
A. specific weight C. column of liquid
B. dead weight D. leverage pressure

98. What is the measure of the quantity of an element or a compound
A mass C. volume
B. mole D. weight

99. Part of a program that makes decisions about the data, whether it is valid or should be sent to the database?
A. client and peers
B. application logic
C. presentation management
D. application layer

100. What is the present worth of cost associated with an asset for an Infinite period of time?
A. Incremental cost C. capitalized cost
B. cost D. annual cost

SET 10

1. Deformation per unit length in the direction of force is known as
   A. strain C. linear strain
   B. lateral strain D. unit strain
2. Which device of computer operation dispenses with the use of keyboard?
   A. Joystick C. Mouse
   B. Light pen D. Touch
3. Article IV, Rule ___ Requirements for examination as a REE
   A. 14 C. 16
   B. 15 D. 17
4. When a force is called a normal force, It Is
   A. The usual force expected given the arrangement of a system
   B. A force that Is perpendicular to the surface of the earth at any given location
   C. A force that is always vertical
   D. A contact force perpendicular to the contact surfaces between two solid objects.
5. What colors are used on signs for lighting equipment?
   A. Black pictogram on yellow background
   B. white Pictogram on a red background
   C. White Pictogram on a green equipment
   D. none of these
6. What type of bond exist predominantly in chlorine sodium chloride crystal
   A. Ionic bond C. covalent bond
   B. metallic bond D. polar bond
7. Which of the following group of elements all form crystal of diamond lattice
   A carbon, silicon, germanium
   B. carbon, boron, aluminum
   C. carbon, cobalt, nickel
   D. lithium, sodium, potassium
8. Which of the following is the ideal Gas Law?
   A. V/T = k C. PV = nRT
   B. $P_1/T_1 = P_2/T_2$ D. V = k × 1/P
9. In molecules of the same composition, what are variations of atomic arrangements known as?
   A. Polymers C. isomers
   B. monomers D. crystal systems
10. The vector which represent the sum of a group of force vectors is called the
    A. Magnitude C. sum
    B. resultant D. force polygon
11. A diagram which shows the variations of the axial load for all sections of the span pf a beam, is called
    A. Bending moment diagram
    B. Thrust diagram
    C. Shear force diagram
    D. Stress diagram
12. The ratio of the lateral strain to the linear strains is
    A modulus of elasticity C. bulk modulus
    B. modulus of rigidity D. Poisson's ratio
13. Otto cycle is also known as
    A. Constant pressure cycle
    B. Constant temperature cycle
    C. Constant volume cycle
    D. Constant entropy cycle
14. What is the most common alloying ingredient in copper?
    A. Iron C. Lithium
    B. Zinc D. Magnesium
15. What refers to the case hardening process by which the carbon content of the steal near the surface of a part is Increased?
    A. carburizing C. normalizing
    B. annealing D. tempering
16. Which of the following conductors is not applicable on wet locations?
    A Type THHN C. Type RHW
    B. Type THW D. Type THWN
17. Which characteristics of a wave changes as the wave travels across a boundary between two different media?
    A. Frequency C phase
    B. period D. speed
18. The thermodynamic statement of conversion of energy of a system is the
    A. first law C. third law
    B. second law D. zeroth law
19. Which of the following is not a function of computer?
    A. Understanding C. processing
    B. controlling D. inputting
20. Change In enthalpy occurs at constant___
    A. Pressure C. temperature
    B. volume D. entropy
21. The first law of thermodynamics is based on which of the following Principles?
    A. Conservation of mass
    B. Action and reaction
    C. Conservation of energy
    D. the entropy temperature relationship
22. Impulse type turbine named after his creator.
    A. Pelton C. Frands
    B. Kaplan D. Propeller
23. Electrodes of rods of iron or steel shall be at least ____ mm In diameter.
    A. 13 C. 20
    B. 16 D. 6
24. At least___mm of three conductor, measured from the point in the box where It emerges from Its raceway shall be left at each outlet, junction and switch point for splices or connection to fixtures or device.
    A. 150 C. 300
    B. 250 D. 500
25. Common cause of failure of termination cable.
    A splicing C. pencilling of insulation
    B. clearance D. none of these
26. A process where a physical mixture of carbide and powdered metal is heated in order to solidify the powder into a single piece.
    A Vitrification C alloying
    B. bonding D. sintering
27. An elevator is pulled by cable at a constant speed. The tension in the cable is equal to
    A. the weight C. thrice the weight
    B. twice the weight D. half the weight
28. What is the form of the moment diagram of a beam supported at both ends and carrying a uniform distributed load?
    A. triangle C parabola
    B. hyperbola D. rectangle
29. Which of the following is not part of the requirements to take RME?
    A. taken 1-yr electrician course and 3-yrs apprenticeship
    B. taken 2-yr electrician course and 3-yrs apprenticeship
    C. taken 3-yr electrician course and 3-yrs apprenticeship
    D. graduate of BSEE
30. What is CPE in RA 7920 stands for?
    A. Continuous Professional Education
    B. Continuing Professional Education
    C. Continuous Professional Educator
    D. Continuing Professional Educator
31. Varignon's theorem is used to determined the___
    A. moment of inertia C. location of centroid
    B. mass moment of inertia D. moment of area
32. Work is measured in the same units as?
    A. force C. mass
    B. momentum D. energy
33. Adiabatic process is?
    A no heat transfer C. irreversible
    B. reversible D. constant pressure
34. The equipment grounding conductor of a branch circuit shall be identified by what color?
    A. green C. white
    B. black D. gray
35. Bernoulli's law for fluid is based on which if the following Principles?
    A. continuity equation
    B. Newton's second low
    C. conservation of energy
    D. conservation of mass
36. Pluto travels from perihelion to aphelion. What Is the work done by gravity on Pluto?
    A. zero C. positive
    B. negative D. undetermined
37. Which of the following is widely used as light metal?
    A. aluminum C. copper
    B. silver D. steel
38. Which is a vector quantity?
    A. time C. work
    B. displacement D. distance

39. Which of the following is NOT a vector quantity?
A displacement C speed
B. acceleration D. momentum

40. Ground Fault Circuit Interrupter (GFCI) must be installed in
I-basement,
II-attachable garage,
III-Detachable garage;
A. I C. II and III
B. I and II D. I,II and III

41. It is the ratio of total number of sustained customers power interruption within given period to the total number of customers served within the same period.
A. SAIFI C. MAIFI
B. SAIDI D. WAIFI

42. Which of the following elements and compounds is unstable in Its pure form?
A. helium, He C. carbon dioxide, C02
B. sodium, Na D. hydrochloric acid, HCI

43. In a kitchen, the minimum number of branch circuit is,
A. 2 C. 1
B. 3 D. 6

44. The derating factor for 3 wines in a conduit is____
A. 1 C. 0. 8
B. 0. 7 D. 0. 9

45. An open system first law should be utilized for all of the following except,
A. a nozzle C. a turbine
B. a piston cylinder device D. a compressor

46. Sec 17 (b) of Art III: Examination and Registration, Qualification of Applicants for Registered Electrical Engineer, states that applicant is at least be___ years of age.
A. 21 C. 24
B. 23 D. 25

47. It is the sum of the Internal energy and PV.
A. entropy C. energy
B. enthalpy D. electric field

48. Heavy water has a mass of
A. 12 C. 20
B. 18 D. 36

49. An external pressure exerted on a fluid is transmitted uniformly throughout the volume of the fluid. This statement is known as___
A. Bernoulli's theorem C. Archimedes Principle
B. Pascal's theorem D. Torricelli's theorem

50. What refers to the point in which bending moment changing sign through a zero value in strength of materials?
A. Zero stress point C. Point of contraflexure
B. Critical point D. Point of inflection

51. In dwelling units at least one receptacle outlet shall be installed in hallways_____meters or more.
A. 4 C. 3
B. 2 D. 1. 8

52. A bar is equivalent to how many pascals?
A. 10^3 C. 10^5
B. 10^4 D. 10^6

53. What law provides an absolute reference point for the *determination* of entropy?
A. Zeroth law of thermodynamics
B. second law of thermodynamics
C. First law of thermodynamics
D. third law of thermodynamics

54. MC cable insulation shall have a maximum operating temperature of not less than____
A. 75 deg C C. 90 deg C
B. 80 deg C D. 60 deg C

55. Steels can be strengthened by all of the following practices, EXCEPT:
A. Annealing
B. Quenching and tempering
C. work hardening
D. grain refinement

56. Which of the following metals has the highest melting point?
A. Copper C. silver
B. gold D. tungsten

57. The term "enthalpy" comes from the Greek "enthalpy" which means
A. Warm C. Heat
B. Hot D. Cold

58. Study of motion with reference to the force which causes the motion is:
A. kinematics C. statics
B. kinetics D. dynamics

59. The inner strand of ACSR Is made of
A. Brass C. copper
B. steel D. lead

60. Charles's law relates to
A. constant volume C. constant pressure
B. constant temperature D. constant entropy

61. The element in the preset Periodic table are arrange according to their
A. atomic numbers C. oxidation states
B. atomic masses D. mass number

62. The term "brass" is very commonly used to *designed* my alloy primarily
A. aluminum and Iron C. copper and aluminum
B. zinc and nickel D. Copper and zinc

63. Which of the following refers to the diffusion of a solvent into a stronger solution in an attempt to equalize the two concentrations?
A. purification C. osmosis
B. electrolysis D. hydrolysis

64. It is a general name to a wide range of alloys containing Iron, carbon, and silicon and to a lesser extent, manganese, phosphorus and sulfur
A. wrought iron C. hard iron
B. cast iron D. smooth iron

65. Which of the following is the busiest bus and the one that carries that carries that actual data?
A. address bus C. control bus
B. data bus D. microscope

66. Which of the following is usually a temporary connection?
A. Bus C. Jumper
B. Tie Wire D. Feeder

67. Which of the following relationship defines enthalpy?
A. H = pv + T C. h = u + p/v
B. h = u + pv D. h = u + p/T

68. The ideal, reversible Camot cycle involve four basic processes. What type of processes are they?
A. All isothermal
B. two adiabatic and two isentropic
C. All adiabatic
D. two isothermal and two isentropic

69. In any collision, which of the following is conserved?
A. Kinetic energy C momentum
B. velocity D. potential energy

70. For hospitals, a general lighting load of _VA/m shall be used.
A. 12 C. 8
B. 24 D. 16

71. A method identifying the location of a given data record by Its name.
A. Tracking C. Signature
B. Linker D. Hashing

72____is an act of sending e-mails or creating web pages that are designed to collect an individual's online bank, credit card or other login information?
A. Malware C. Cracking
B. Hacking D. Phishing

73. Which of the following wires is applicable for underground service entrance conductors?
A. type THWN C. type MI
B. type UF D. type USE

74. Constantan, an alloy used in thermocouples, is an alloy of
A. copper and tin C. copper and iron
B. copper and zinc D. copper and nickel

75. In Joule cycle, heat is rejected at constant
A. Volume C. Temperature
B. Pressure D. Entropy

76. When a part is constrained to move when heated, it develops
A. principal stress C. compressive stress
B. tensile stress D. shear stress

77. Which group of the periodic table is known as the alkali metals?
A. Group 1 C. Group 3
B. Group 2 D. Group

78. Which method is used to connect a remote computer?
A. Device C. Diagnostic
B. Dial up D. Logic circuit

79. The ratio of the velocity of light in vacuum to its velocity of another material is
A. Specific Index C. Refractive Index
B. Poisson's Rato D. Module ratio of Elasticity

80. What is another term for "acid-test ratio?
A. Quick ratio C. Current ratio
B. Price-earning ratio D. Profit margin ratio

81. The chemical name for baking soda is
A. Sodium bicarbonate C. Sodium chloride
B. Sodium sulphate D. Sodium carbonate

82. The most malleable metal is
A. Platinum C. Iron
B. Siver D. Gold

83. A modem is connected to
A. telephone line C. a printer
B. a keyboard D. a monitor

84. The alternate banking Delivery Channels are:
A. E-banking C. ATM
B. mobile banking D. SMS banking

85. Sec. 14 (a) (iii) of Art. III. Examination and registration, states the prior to commending of work the foreign professional shall secure special permit from the_____
A. Commission C. POEA
B. IIEE D. Bureau of Immigration

86. The basic unit of a worksheet into which you enter obtain Excel is called a
A. Tab C. box
B. cel D. range

87. Which Is the longest key on the keyboard board?
A. Space bar C. Function key
B. Enter key D. Shift Key

88. The first law of thermodynamics is based on which of the following Principles?
A. conservation of mass
B. the enthalpy entropy relationship
C. action-reaction
D. conservation of energy

89. Which of the following DO statements counts J from 1 to 11in increments of 2?
A. DO 5 J=1,2,11 C. DO 5 J=1. 11.2
B. DO 5,J=1,11,2 D. DO 5 J=1,11,2

90. Which of the following statements is FALSE for an ideal gas?
A. The molecules behave as solid spheres of finite radius.
B. pV = nRT
C. Collisions between gas molecules are perfectly elastic and result in no decrease in kinetic energy.
D. No attractive forces exist between the molecules.

91. Which of the following statements is FALSE?
A. The time rate of change of the angular momentum about a fixed point is equal to the total moment of the external forces acting on the system about the point.
B. The coefficient of restitution can be less than zero.
C. The frictional force always acts to resist motion.
D. Momentum is conserved during elastic collisions.

92. Which of the following is qualified as the Chairman of the Continuing Professional Development Council for Electrical Engineering?
A. Representative from Association of Professional Organization
B. PRBEE board member
C. Representative from Academy
D. PRC Commissioner

93. Which of the following devices merges signals?
A. Oscillator C. Demodulator
B. Multiplexer D. Demultiplexer

94. A type of computer memory circuit that holds data, program Instructions, and the operating system while the computer Is on.
A. Random access memory
B. Read-only memory
C. Volatile memory
D. CMOS

95. Common cause of failure of termination of cable.
A. splicing
B. clearances
C. penciling of Insulation
D. none of these

96. Why do ideal gas exerts pressure on the wall of its container?
A. Because of the force of the weight of the molecules
B. Because the molecules collide the wall Inelastically
C. Because the molecules collide the wall continuously
D. Because the molecules collide each other

97. The equation which states that the total energy in the steady flow of a frictionless and incompressible fluid is constant
A continuity equation
B. Reynolds equation
C. Bernoulli's equation
D. general energy equation

98. DMC stands for_____
A. Distribution Market committee
B. Distribution Management commlssion
C. Distribution Management committee
D. Distribution Market commission

99. Vacancies of the board of Electrical Engineering shall be filled by the from the list of recommendees selected by the PRC commissioner.
A. Chairman of the Board
B. President of the Philippines
C. IIEE
D. PRC Commissioner

100. The usual function of a disconnect switches in high voltage circuits is to
A. open the circuit in the event of an overload
B. maintain continuity of service
C. Isolate from energized buses, equipment which are not in service
D. open or dose the circuit under load

*Reference: CoachG, for Engr2025 ONLY!